

Community Sports Facilities Booking System

Project Documentation

A web-based prototype for a municipal sports council, built with ASP.NET Core MVC and SQL Server.

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Module: Data and Web Application Development

Coursework: Community Sports Facilities Booking System

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1. Introduction

The Community Sports Facilities Booking System is a web-based application built for a municipal sports council. It allows members of the public to find out about local sports facilities such as tennis courts, swimming pools, sports halls, football pitches, athletics tracks and cricket and volleyball grounds, and lets registered members book those facilities online.

The system was developed as a university coursework project. It is a working prototype rather than a full commercial product, but it implements a complete and genuinely functional set of features, including a proper database design, a secure member account system, real-time booking with conflict prevention, and an administration area for managing the system.

This document describes the purpose of the system, how it was designed and built, the technologies used, how to set it up and run it, its main features and modules, and the testing that was carried out on the finished application.

2. Problem Statement

A municipal sports council wants to make it easier for the local community to find and use public sports facilities. At present, booking a facility and finding out what is available is not centralised, which can lead to scheduling conflicts, wasted trips, and facilities being under-used because people do not know they exist or whether they are free at a given time.

The council needs an online system that lets the public browse and search for facilities, lets registered members book a facility for a specific date and time without double-booking, lets members share feedback through reviews, and gives the council a way to see and manage this activity. Guests who are not yet members should still be able to see basic information, read reviews, register, and get in touch with a question, without needing to create an account first.

3. Objectives

The main objectives of the project are:

- Design a normalised relational database that correctly represents members, sports, facilities, bookings, reviews and inquiries.
- Build a web application, using ASP.NET Core MVC with C#, that implements the case study's Guest and Member features.
- Allow a member to register with personal details and preferred sports, sign in securely, search for facilities, book a facility for a specific date and time, and leave a review after using it.
- Allow a guest (a visitor without an account) to browse and search facilities with limited detail, read reviews, register to become a member, and send an inquiry.
- Prevent double-booking of the same facility for an overlapping date and time, using database-level protection rather than relying on the application alone.
- Add simple administration tools so that the facilities, sports, members, bookings, reviews and inquiries in the system can be managed and reviewed.
- Test the finished system thoroughly, using both automated tests and real, manually driven tests of the running application, and document the results honestly.

4. Technologies and Tools Used

The technologies used follow the coursework's required stack: ASP.NET Core with C# for the web application, and SQL Server for the database.

4.1 Backend

- ASP.NET Core MVC (.NET 10) - the web application framework, using the Model-View-Controller pattern.
- C# - the programming language used for all server-side code.
- Entity Framework Core 10 - used to read and write data between the C# code and the SQL Server database.
- Microsoft SQL Server - the relational database that stores all application data.
- ADO.NET (Microsoft.Data.SqlClient) - used to call a SQL Server stored procedure directly for booking creation, in addition to Entity Framework Core.

4.2 Frontend

- Razor views (.cshtml) - used to generate the HTML pages on the server.
- HTML5 and CSS3 - the page structure and styling.
- Bootstrap 5 - the CSS framework used for layout, forms, buttons and the responsive navigation menu.
- Bootstrap Icons - the icon set used throughout the interface.
- JavaScript and jQuery, with jQuery Unobtrusive Validation - used for client-side form validation messages and small interactive features such as the quick time-slot picker.

4.3 Tools

- Oracle SQL Developer Data Modeler - used to build the entity-relationship diagram and the physical database model, as required by the coursework.
- SQL Server Management Studio and sqlcmd - used to create the database, run the SQL scripts, and inspect data during development.
- Git - used for version control of the source code.
- xUnit - the automated testing framework used for the C# test suite.
- A browser automation tool (Playwright, driving a real installed Chrome browser) - used during this documentation exercise to actually operate the running application and capture genuine screenshots of each test case.

5. System Requirements

To run this project, the following are required:

- Windows 10 or later (the project was developed and tested on Windows 11).
- .NET 10 SDK.
- Microsoft SQL Server (Developer or Express edition is sufficient), running locally or reachable over the network.
- A modern web browser, such as Microsoft Edge, Google Chrome or Firefox.
- Visual Studio 2022 or Visual Studio Code (optional, but recommended for editing and debugging).

- An internet connection the first time the project is built, so that NuGet can download the Entity Framework Core packages.

6. Project Setup and Installation

The steps below set up the database and the web application from a copy of the project source code.

6.1 Database setup

The SQL scripts in the database folder must be run once, in order, against a SQL Server instance. Each script is a plain .sql file and can be run using SQL Server Management Studio or sqlcmd.

- 01_CreateDatabase.sql - creates the CommunitySportsBookingDB database.
- 02_CreateTables.sql - creates the 8 application tables with their keys and constraints.
- 03_Indexes.sql - adds the indexes used for search and availability queries.
- 04_BookingOverlapProtection.sql - creates the usp_CreateBooking stored procedure and the trg_Booking_PreventOverlap trigger that prevent double-booking.
- 05_SeedData.sql - inserts sample members, sports, facilities, bookings, reviews and inquiries.
- 06_TestQueries.sql - a set of sample SELECT queries used to demonstrate and check the data.
- 07_AdminSupport.sql - adds the IsAdmin column used by the administration area.
- 08_BookingCancellation.sql - adds booking cancellation support (IsCancelled, CancelledDate) and updates the overlap checks to ignore cancelled bookings.

6.2 Web application setup

1. Open a command prompt in the project folder (the folder containing CommunitySportsBooking.slnx).
2. Check the connection string in src/CommunitySportsBooking.Web/appsettings.json. By default it points to a local SQL Server instance using Windows authentication:
Server=localhost;Database=CommunitySportsBookingDB;Trusted_Connection=True;TrustServerCertificate=True;. Change this if a different server or authentication method is used.
3. Run "dotnet restore" to download the required NuGet packages.
4. Run "dotnet build" to compile the solution and confirm there are no errors.
5. Run the seed hash update command once: "dotnet run --project src/CommunitySportsBooking.Web -- --update-seed-hashes". This replaces the placeholder password values in the seed data with real, securely hashed passwords.
6. Run the admin account seed command once: "dotnet run --project src/CommunitySportsBooking.Web -- --seed-admin-account". This creates the dedicated administrator sign-in (email "admin", password "admin").

7. How to Run the Project

Once the database and application have been set up, the project can be started with:

```
dotnet run --project src/CommunitySportsBooking.Web --launch-profile http
```

The application starts on http://localhost:5233. Open this address in a web browser.

7.1 Demo accounts

The following accounts already exist in the sample data and can be used to try out the system:

- Member account - Email: `alice.johnson@example.com`, Password: `Demo@Pass123`. Four other sample members exist with the same password pattern (`ben.carter@example.com`, `chloe.davies@example.com`, `daniel.evans@example.com`, `emma.foster@example.com`).
- Administrator account - Email: `admin`, Password: `admin`. Signing in with this account leads straight to the Admin area.

New visitors can also register their own member account from the Register page.

8. System Features Overview

The system supports three types of user: Guest (not signed in), Member (a registered, signed-in user), and Administrator (a member with management rights).

8.1 Guest features

- View the home page, which introduces the sports council and shows popular sports and a sample of active facilities.
- Browse the full list of active facilities and view a facility's details.
- Run a restricted search of facilities by type/sport and location, without seeing live date/time availability.
- Search and read member reviews of facilities.
- Register to become a member, including choosing preferred sports.
- Send an inquiry (a contact message) to the sports council.

8.2 Member features

- Sign in and out of a personal account.
- Search facilities with a specific date and time window and see real-time availability.
- Book a facility for a chosen date and time, protected against double-booking.
- View "My Bookings", grouped into Upcoming, Completed and Cancelled.
- Cancel an upcoming booking that has not yet started.
- Submit a review, with a rating and comment, for a completed booking.
- View and edit personal profile details, and manage preferred sports.

8.3 Administrator features

- View a dashboard of live statistics about the whole system.
- Search, view and activate/deactivate member accounts.
- Add, edit, and view sports, and delete a sport that is not in use.
- Add, edit, activate/deactivate and view facilities, assign the sports a facility supports, and delete a facility with no booking history.
- Search and filter all bookings across every member, view booking details, and cancel a booking on a member's behalf.
- View and update the status of guest inquiries (New, Reviewed, Closed).

- View all member-submitted reviews across every facility.

9. Module and Component Explanation

The application follows the standard ASP.NET Core MVC structure: Controllers handle web requests, Models and ViewModels represent data, Views display HTML pages, and a small number of Services hold shared business logic so that it is written once and reused wherever it is needed.

9.1 Controllers

- HomeController - the home page, the generic error page, and the branded page shown for HTTP error codes such as 404.
- AccountController - member registration, login, logout, and the "access denied" page.
- ProfileController - viewing and editing a member's own profile and preferred sports.
- FacilityController - browsing facilities, viewing facility details, and the shared Guest/Member facility search.
- BookingController - creating a booking, viewing "My Bookings", and cancelling a booking.
- ReviewController - submitting a review for a completed booking, and the public review search.
- InquiryController - the guest contact/inquiry form.
- Areas/Admin/Controllers - a separate set of controllers (Dashboard, Members, Sports, Facilities, Bookings, Inquiries, Reviews) that only an administrator can reach, kept in their own "Admin" area of the project.

9.2 Services

A small number of static service classes hold logic that more than one controller needs, so that the same rule is never implemented twice:

- BookingService - creates a booking by calling the usp_CreateBooking stored procedure, and works out whether a booking is Upcoming, Completed or Cancelled.
- FacilityAvailabilityService - checks whether a facility is free for a requested date and time window (used by both search and booking).
- SportsPreferenceService and FacilitySportsService - save a member's preferred sports or a facility's supported sports, working out which rows to add and remove.
- ReviewService - looks up reviews for a facility or across a search.
- LoginAttemptTracker - keeps track of recent failed login attempts in memory, and temporarily locks out an email address after 5 failures in a row.
- ImageResolver - chooses a suitable picture or icon for a facility or sport based on its type or name.

9.3 Data access

Entity Framework Core is used to read and write the 8 database tables (Member, Sport, MemberSport, Facility, FacilitySport, Booking, Review, Inquiry) through an AppDbContext class. The database schema itself is owned entirely by the SQL scripts described in Section 6.1 - Entity Framework Core is only used to map to that existing schema; it never creates or changes the database structure automatically. Booking creation is the one exception that goes around Entity

Framework Core's normal insert path: it calls the `usp_CreateBooking` stored procedure directly, so that the database-level locking and overlap checks inside that procedure are always used.

9.4 Authentication and Authorization

Member accounts and passwords are stored in the application's own `Member` table, not in a separate identity system. Passwords are never stored as plain text: they are hashed using ASP.NET Core's `PasswordHasher` class before being saved, and a login only succeeds if the submitted password hashes to the same value. After a successful login, the browser is given a secure cookie that identifies the member for future requests. An administrator is simply a member whose `IsAdmin` flag is set to true; when such a member logs in, an extra "Admin" role is added to their session, and every controller inside the Admin area requires that role before it will respond.

10. Main User Flows

10.1 Guest registration flow

1. A visitor opens the Register page.
2. They enter their name, email, phone number, address, city and a password, and optionally select one or more preferred sports.
3. The system checks that all required fields are filled in correctly and that the email is not already registered.
4. If everything is valid, a new `Member` record is created, the password is hashed and stored, the chosen sports are saved, and the visitor is signed in automatically.

10.2 Member search and booking flow

1. A signed-in member opens Search Facilities and enters a sport/type, a location, and optionally a date and time window.
2. The system returns matching active facilities. If a date and time were given, each result also shows whether that facility is free for that window.
3. The member opens a facility and clicks "Book This Facility", then confirms a date and start/end time.
4. The system checks that the date is not in the past and that the start time is before the end time, then checks whether the slot is still free.
5. If the slot is free, the booking is created through the `usp_CreateBooking` stored procedure, which uses a database lock and a safety-net trigger to guarantee that two people cannot book the same facility for an overlapping time, even if they submit at the same moment.
6. The member is shown their new booking under "My Bookings".

10.3 Review submission flow

1. A member opens "My Bookings" and finds a booking in the Completed section that has not yet been reviewed.
2. They click "Leave a Review", choose a star rating from 1 to 5, and write a comment.
3. The system checks that the booking belongs to that member, that it is genuinely completed, and that it has not already been reviewed.
4. If all checks pass, the review is saved and linked to that booking.

10.4 Guest inquiry flow

1. A visitor opens the Contact Us page.
2. They enter their name, email, a subject and a message.
3. The system checks that all fields are filled in and that the email looks valid.
4. The inquiry is saved with a status of "New" so that an administrator can review it later.

10.5 Administrator management flow

1. An administrator signs in and is taken straight to the Admin Dashboard.
2. From the sidebar, they can open Members, Sports, Facilities, Bookings, Reviews or Inquiries.
3. Each management page lets the administrator search or filter, view details, and carry out the actions appropriate to that area (for example, deactivating a member, cancelling a booking, or updating an inquiry's status).
4. Actions that would remove information the system depends on, such as deleting a sport that is still used by a facility, or deleting a facility that already has bookings, are blocked with a clear explanation.

11. Validation and Error Handling

Every form in the system is validated on the server before any data is saved, in addition to the immediate feedback shown by client-side JavaScript. The main validation and business rules are summarised below.

11.1 Field-level validation

Form	Field	Rule
Register	First name, Last name	Required, 1-50 characters.
Register	Email	Required, must be a valid email format, must not already be registered.
Register	Phone	Required, 7-20 characters, digits/spaces/+/-/() only.
Register	Password	Required, at least 8 characters, must contain at least one letter and one digit.
Register	Confirm Password	Required, must match Password.
Login	Email, Password	Both required. The Email field also accepts the non-email administrator identifier ("admin").
Profile Edit	Phone	Same rule as registration. Email cannot be changed once registered.
Booking	Date	Required, cannot be in the past, cannot be more than 90 days ahead.
Booking	Start/End Time	Both required, start time must be before end time.
Review	Rating	Required, whole number from 1 to 5.

Form	Field	Rule
Review	Comment	Required, 1-1000 characters.
Inquiry	Name, Subject, Message	Required, with a maximum length for each field.
Inquiry	Email	Required, must be a valid email format.

11.2 Business-rule validation

- Booking overlap prevention - a facility cannot be booked twice for the same or an overlapping time on the same day. This is enforced in three layers: an application-level check for quick feedback, a database lock inside the `usp_CreateBooking` stored procedure that closes the race condition between two simultaneous requests, and a database trigger that acts as a final safety net regardless of how a booking row was created.
- Login lockout - after 5 failed login attempts for the same email address, further attempts are blocked for 5 minutes. This applies whether or not the email belongs to a real account, so the lockout itself cannot be used to guess which emails are registered.
- Generic login errors - a wrong password, an unrecognised email, and a deactivated account all show the same message ("Invalid email or password."), so a visitor cannot use the error message to work out which of these is true.
- Guest search restriction - if a visitor who is not signed in tries to send date/time search parameters directly (bypassing the search form), the server ignores them rather than trusting them, so a guest can never see live availability without logging in.
- Ownership checks - a member can only view, cancel or review their own bookings. Requests for a booking that does not exist or does not belong to the current member are treated the same way (a standard "not found" response), so the error message never reveals whether the booking exists at all.
- Referential protection - the database refuses to delete a sport that is still used by a facility or preferred by a member, and refuses to delete a facility that already has bookings. The administration pages detect this in advance and explain it, rather than showing a technical database error.

11.3 Error pages

If a page is not found, or another HTTP error occurs, the application shows a branded error page instead of a plain server error. Unexpected server errors are caught and shown as a friendly error page rather than exposing technical details to the user.

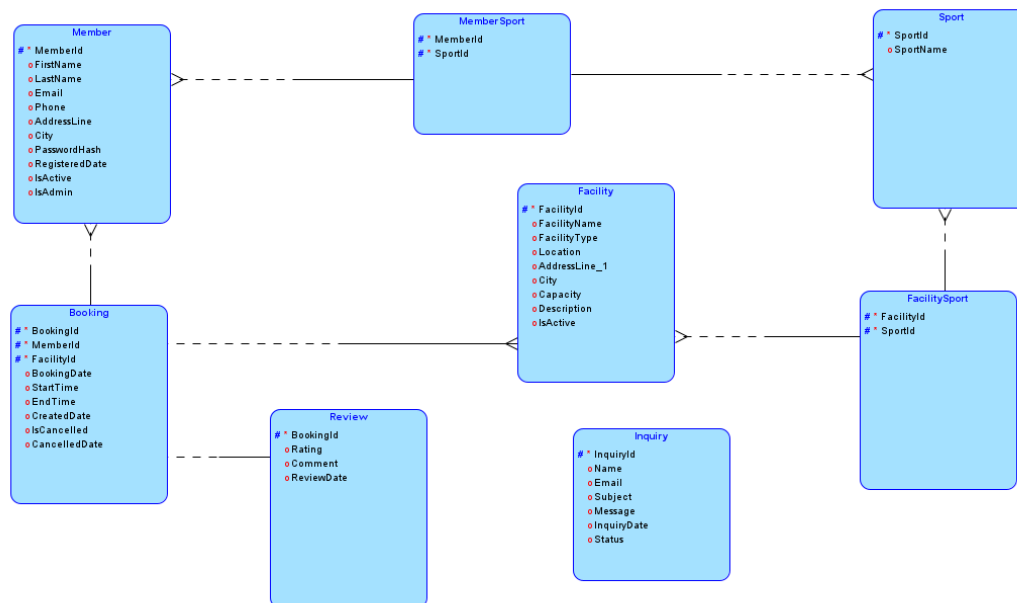
12. Database Design

The database has 8 tables: Member, Sport, MemberSport, Facility, FacilitySport, Booking, Review and Inquiry. MemberSport and FacilitySport are bridge tables that record a many-to-many relationship (which sports a member prefers, and which sports a facility supports) without repeating data.

An entity-relationship diagram and a physical database model were produced separately using Oracle SQL Developer Data Modeler, as required by the coursework. Both are shown below, followed by the same design described as text tables.

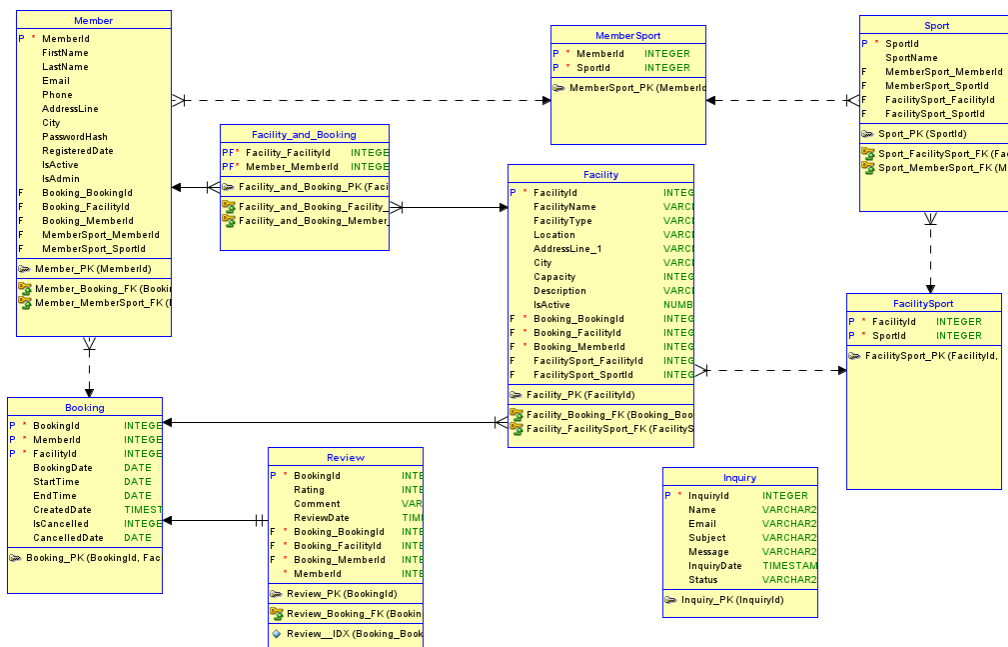
12.1 Entity-Relationship Diagrams

The logical model below shows the entities and their relationships at a conceptual level.



Logical entity-relationship model, produced in Oracle SQL Developer Data Modeler.

The relational (physical) model below shows the same design engineered into tables, columns, data types and keys for the SQL Server RDBMS site.



Relational (physical) model, produced in Oracle SQL Developer Data Modeler.

12.2 Table structures

Member

Column	Type	Notes
MemberId	INT, identity	Primary key.
FirstName, LastName	NVARCHAR(50)	Required.
Email	NVARCHAR(256)	Required, unique.
Phone	NVARCHAR(20)	Required.
AddressLine, City	NVARCHAR	Required.
PasswordHash	NVARCHAR(MAX)	Hashed password, never plain text.
RegisteredDate	DATETIME2	Defaults to the current date and time.
IsActive	BIT	Defaults to 1 (active). Set to 0 to deactivate a member.
IsAdmin	BIT	Defaults to 0. Set to 1 to give administrator rights.

Sport

Column	Type	Notes
SportId	INT, identity	Primary key.
SportName	NVARCHAR(50)	Required, unique (e.g. "Tennis", "Swimming").

MemberSport

Column	Type	Notes
MemberId	INT	Part of the primary key. Foreign key to Member, cascades on delete.
SportId	INT	Part of the primary key. Foreign key to Sport.

Facility

Column	Type	Notes
FacilityId	INT, identity	Primary key.
FacilityName	NVARCHAR(100)	Required.
FacilityType	NVARCHAR(50)	Required (e.g. "Tennis Court", "Sports Hall").
Location	NVARCHAR(100)	Required. Area name used for search.
AddressLine, City	NVARCHAR	Required.
Capacity	INT	Optional.
Description	NVARCHAR(1000)	Optional.
IsActive	BIT	Defaults to 1. Inactive facilities are hidden from search and booking.

FacilitySport

Column	Type	Notes
FacilityId	INT	Part of the primary key. Foreign key to Facility, cascades on delete.
SportId	INT	Part of the primary key. Foreign key to Sport.

Booking

Column	Type	Notes
BookingId	INT, identity	Primary key.
MemberId	INT	Foreign key to Member. Never deleted, to preserve history.
FacilityId	INT	Foreign key to Facility. Never deleted, to preserve history.
BookingDate	DATE	Required. Cannot be in the past (application-enforced).
StartTime, EndTime	TIME	Required. Start must be before End (database constraint).
CreatedDate	DATETIME2	Defaults to the current date and time.
IsCancelled	BIT	Defaults to 0. Set to 1 when a booking is cancelled.
CancelledDate	DATETIME2, optional	Set when a booking is cancelled.

Review

Column	Type	Notes
BookingId	INT	Primary key and foreign key to Booking. This design means a booking can have at most one review.
Rating	TINYINT	Required, 1 to 5.
Comment	NVARCHAR(1000)	Required.
ReviewDate	DATETIME2	Defaults to the current date and time.

Inquiry

Column	Type	Notes
InquiryId	INT, identity	Primary key.
Name, Email, Subject, Message	NVARCHAR	Required.
InquiryDate	DATETIME2	Defaults to the current date and time.
Status	NVARCHAR(20)	Defaults to "New". Must be "New", "Reviewed" or "Closed".

12.3 Design notes

- The Review table's primary key is the same column as its foreign key to Booking (BookingId). This is a deliberate design choice: it guarantees that a booking can never have more than one review, without needing a separate rule to enforce it.
- Review does not store the member or facility directly. Both can always be found by following the booking it belongs to, which avoids storing the same information in two places.
- MemberSport and FacilitySport have no extra column beyond the two foreign keys that make up their primary key. This is a standard way to represent a many-to-many relationship and automatically prevents the same preference or support relationship being saved twice.
- A booking's Member and Facility are never deleted from the database, even if the member is deactivated or the facility is later removed from search, so that booking history is preserved.
- No BookingStatus column exists. Whether a booking is "Completed" is worked out from its date and end time compared to the current time, so it is always accurate and never goes out of date.

12.4 Concurrency and booking safety

Booking two members into the same facility for an overlapping time is prevented using three layers of protection, so that the system is safe even if two people try to book the same slot at exactly the same moment:

- Application check - before saving, the web application checks whether the slot looks free, so the member gets a fast, friendly message if it obviously is not.
- Database lock - the usp_CreateBooking stored procedure takes an exclusive lock for that facility and date before checking availability and inserting the booking, so that two simultaneous requests cannot both pass the check at the same time.
- Database trigger - a trigger on the Booking table checks for overlaps after any insert or update and cancels the change if one is found, as a final safety net that applies no matter how the row was written.

13. External Services and Data Access

This project does not call any third-party web API or external online service. All data is stored in the local SQL Server database described in Section 12.

The one notable piece of server-side logic beyond ordinary Entity Framework Core reads and writes is the usp_CreateBooking stored procedure. Rather than inserting a booking row directly through Entity Framework Core, the application calls this stored procedure using a direct SQL command, so that the database-level locking described in Section 12.3 is always used for every booking, with no way to accidentally bypass it.

14. Testing

Testing was carried out in two ways: an automated test suite that checks the application's code and database behaviour, and manual functional testing of the real, running application through an actual web browser.

14.1 Automated testing

The project includes an xUnit test suite (in tests/CommunitySportsBooking.Tests) with 185 automated tests. These tests run against a real SQL Server database, not a simulated one, so that database constraints and stored procedures are genuinely exercised rather than assumed. The suite covers, among other things: reading and writing all 8 tables, member registration and login (including the administrator sign-in), profile and sports-preference updates, facility search and booking (including overlap rejection and a genuine concurrent-booking test), booking cancellation, review submission, guest inquiries, and every Admin area feature (dashboard, member/sport/facility/booking/inquiry/review management).

The automated suite was run again as part of preparing this document. The result was:

dotnet test: Passed! - Failed: 0, Passed: 185, Skipped: 0, Total: 185.

14.2 Manual functional testing

In addition to the automated suite, the running application was operated directly through a real web browser to check that the finished system actually behaves correctly from a user's point of view, and to capture genuine evidence of that behaviour. This was done using a browser automation tool driving an installed Chrome browser against the application running on <http://localhost:5233>, connected to the same SQL Server database used for normal development.

42 test cases were carried out this way, covering the Guest, Member and Administrator modules, including normal ("positive") use, invalid input ("negative"/validation) cases, and error-handling and access-control cases. Every test case below was actually executed against the running application; none of the results were assumed or invented. A real screenshot was captured for each test case that produces a visible page result (40 of the 42; the remaining 2 reuse the same redirect mechanism already shown with a screenshot elsewhere and are explained in the notes for that test case).

Two pieces of temporary test data were created specifically to demonstrate certain Admin features (a sport named "Table Tennis" and a facility named "Hillcrest Multi-Sport Hall", used in TC36 and TC38). Both were removed again after their screenshots were captured, so that the reference data used for grading is unchanged from the original sample data. This is noted individually against those two test cases.

15. Test Cases and Results

Each test case below shows the feature being tested, the scenario, what was needed before the test, the exact steps and input used, the expected result, the actual result observed when the test was run, its pass/fail status, and a reference to the supporting screenshot.

15.1 Guest Module

TC01 - Home page loads and shows facility and sport information

Test Case ID	TC01
Feature / Module	Home Page (Guest)
Test Scenario	Home page loads and shows facility and sport information
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Open the application in a web browser.

	2. Observe the home page content.
Test Input / Data	None (navigate to the home page URL).
Expected Result	The home page loads successfully and shows a welcome banner, a "Popular Sports" section, and a "Featured Facilities" section with the number of active facilities.
Actual Result	Home page rendered with Popular Sports and Featured Facilities sections.
Test Status	Pass
Evidence	Screenshot: TC01_Home_Page.png

Find and book sports facilities easily.

The Community Sports Council supports local sports programs and manages a network of public facilities available to every resident.

[Find Facilities](#)

[Browse with Availability](#)

Popular Sports



Athletics



Badminton



Basketball



Cricket



Soccer



Swimming



Tennis

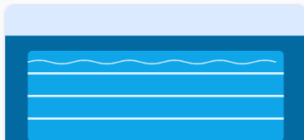


Volleyball

Featured Facilities

7 active facilities currently available to book.

[View all facilities →](#)



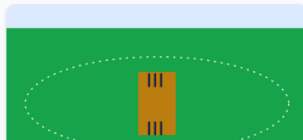
Central Community Pool

City Centre, Springfield

Swimming Pool

Swimming

[Details](#)



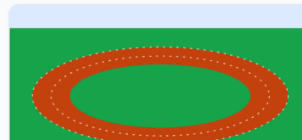
Eastfield Cricket Ground

Eastfield, Springfield

Cricket Ground

Cricket

[Details](#)



Northgate Athletics Track

Northgate, Springfield

Athletics Track

Athletics Cricket

[Details](#)

How It Works



Search

Browse or search facilities by sport and location.



Choose a facility

View details, sports supported, and full facility information.



Select a time

Check real-time availability for the date and time you want.



Confirm

Confirm your booking instantly — no waiting for approval.

Why Book With Us



Easy booking

A simple, guided flow from search to confirmation.



Real-time availability

See exactly which time slots are free before you book.



Multiple sports

8 sports across our facility network.



Secure member accounts

Your bookings and profile are private to you.

[Become a Member](#)

Register to search facilities by date and time, and make bookings.

[Register](#)

NAVIGATE

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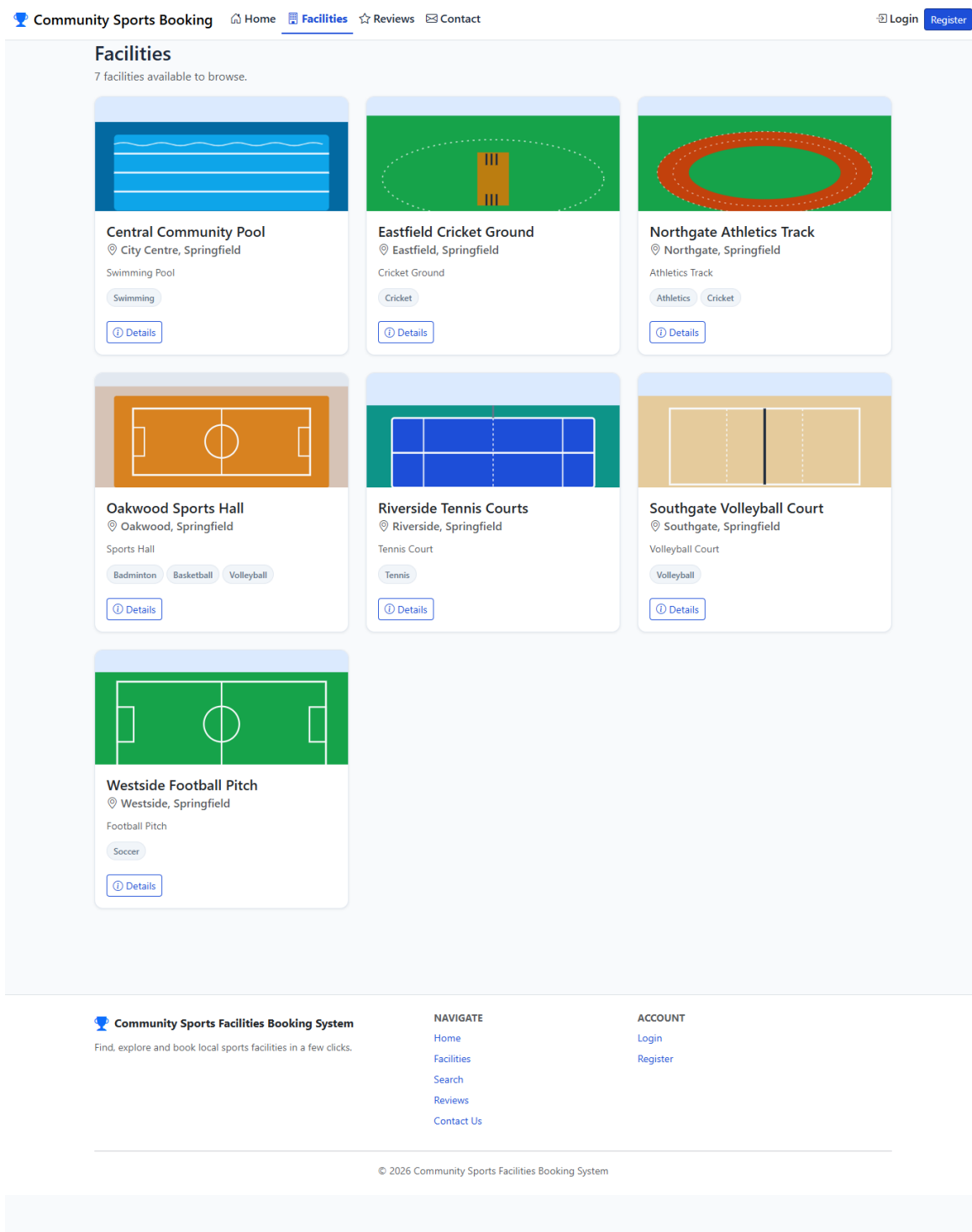
ACCOUNT

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Home page showing popular sports and featured facilities for a guest visitor.

TC02 - Guest browses the list of active facilities

Test Case ID	TC02
Feature / Module	Facility Browsing (Guest)
Test Scenario	Guest browses the list of active facilities
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Click "Facilities" in the navigation bar. 2. Review the list of facilities shown.
Test Input / Data	None.
Expected Result	Only active facilities are listed. The inactive facility ("Old Mill Badminton Courts") must not appear.
Actual Result	Active facilities listed; inactive "Old Mill Badminton Courts" correctly excluded.
Test Status	Pass
Evidence	Screenshot: TC02_Guest_Facility_Browse.png



Facility list page showing only active facilities.

TC03 - Guest views a facility details page

Test Case ID	TC03
Feature / Module	Facility Details (Guest)
Test Scenario	Guest views a facility details page
Preconditions	The application is running. The user has not logged in (Guest).

Test Steps	1. Open the "Riverside Tennis Courts" facility from the facility list. 2. Review the details page.
Test Input / Data	FacilityId = 1 (Riverside Tennis Courts).
Expected Result	The page shows facility information and existing reviews. Because the visitor is not logged in, a "Sign in to Book" button is shown instead of a booking button.
Actual Result	Facility details rendered with "Sign in to Book" prompt and Reviews section.
Test Status	Pass
Evidence	Screenshot: TC03_Guest_Facility_Details.png

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Riverside Tennis Courts

Tennis Court ★ 5 (1 review)

Tennis

ABOUT THIS FACILITY

Two outdoor hard courts by the river.

LOCATION

12 High Street, Springfield
Riverside

CAPACITY

4 people

Reviews

[Browse all reviews →](#)

Alice J.

★★★★★

Excellent courts, well maintained and the surface was in great condition.

30 Aug 2026

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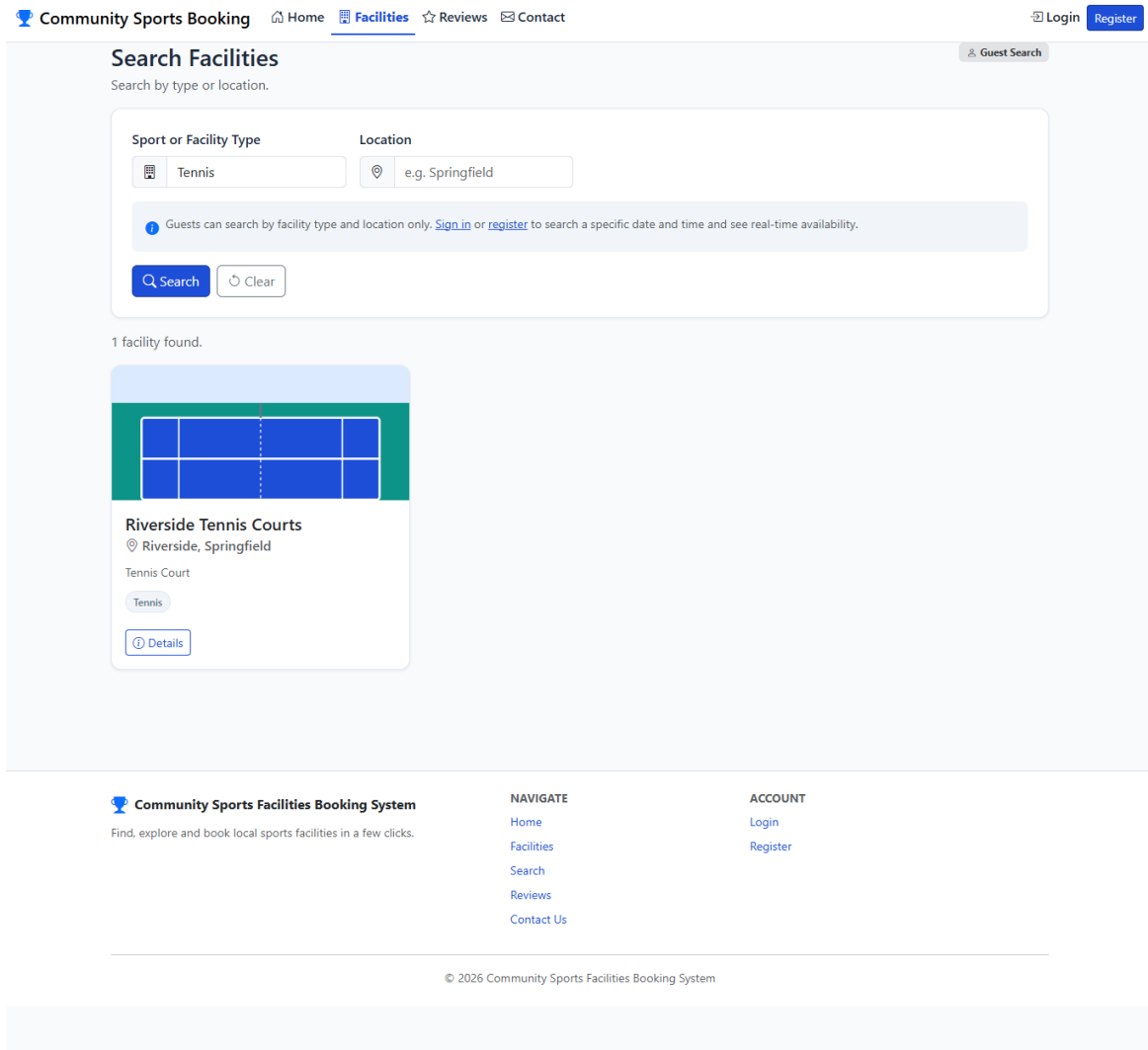
[Register](#)

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Facility details page for a guest, with reviews visible and a sign-in prompt instead of a booking button.

TC04 - Guest performs a restricted facility search (no date or time)

Test Case ID	TC04
Feature / Module	Restricted Search (Guest)
Test Scenario	Guest performs a restricted facility search (no date or time)
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Open the Search Facilities page. 2. Confirm that no date or time fields are shown. 3. Enter "Tennis" in the search box and submit the search.
Test Input / Data	Facility type / sport = "Tennis".
Expected Result	The search form does not show date or time fields for a guest. The search returns "Riverside Tennis Courts" as a match.
Actual Result	Date/time fields absent (found 0); search for "Tennis" returned matching facility.
Test Status	Pass
Evidence	Screenshot: TC04_Guest_Restricted_Search.png



Guest search form with only type and location fields, showing a matching result.

TC05 - Guest searches facility reviews without logging in

Test Case ID	TC05
Feature / Module	Review Search (Guest)
Test Scenario	Guest searches facility reviews without logging in
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Open the Reviews page. 2. Enter "Riverside" as the facility name and submit.
Test Input / Data	Facility name = "Riverside".
Expected Result	Reviews for "Riverside Tennis Courts" are displayed without requiring the visitor to log in.
Actual Result	Review(s) for "Riverside Tennis Courts" returned without requiring authentication.
Test Status	Pass

Evidence	Screenshot: TC05_Guest_Review_Search.png
----------	--

Community Sports Booking Home Facilities **Reviews** Contact Login Register

★ Facility Reviews

Read what other members think of our facilities. No account needed.

Facility Name

Facility Type

Search
Clear

1 review found.

Riverside Tennis Courts · Tennis Court
30 Aug 2026

★★★★★

Excellent courts, well maintained and the surface was in great condition.

— Alice J.

[View facility —](#)

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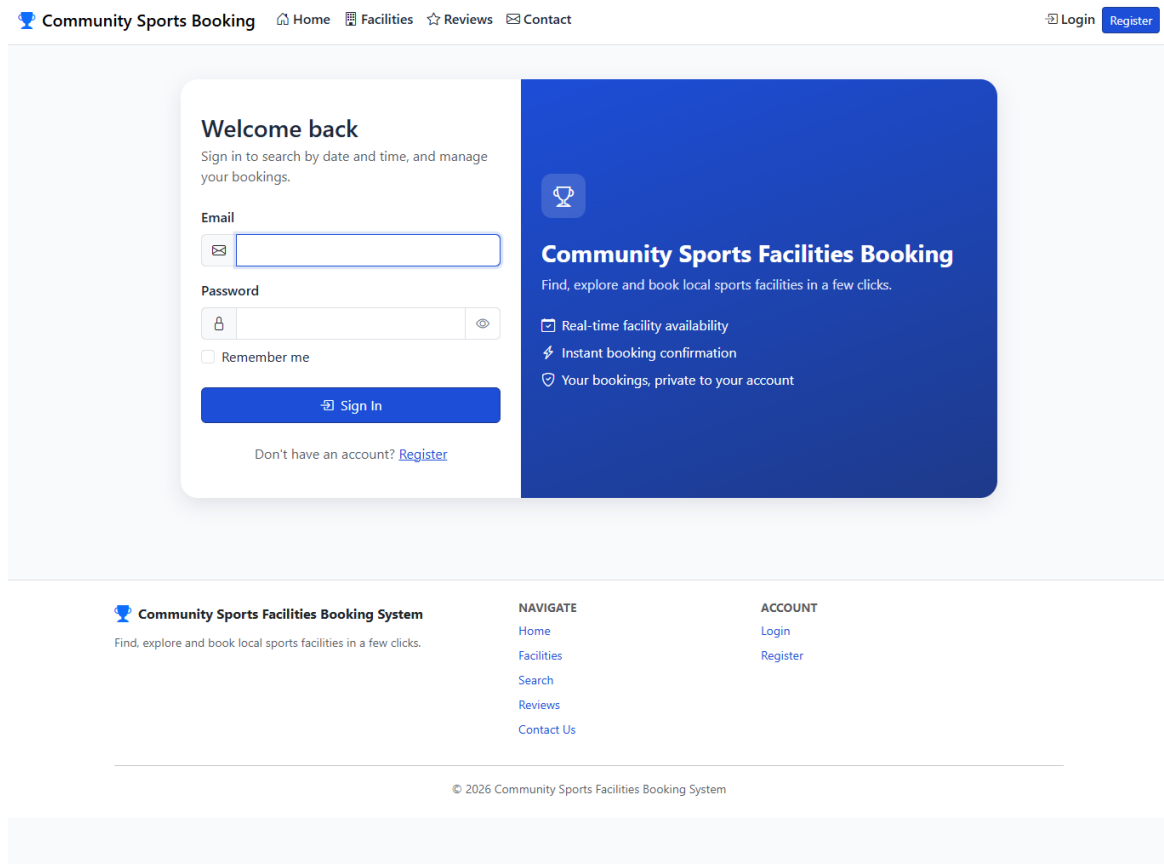
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Guest review search results for Riverside Tennis Courts.

TC06 - Guest is blocked from the booking page and redirected to Login

Test Case ID	TC06
Feature / Module	Access Control (Guest)
Test Scenario	Guest is blocked from the booking page and redirected to Login
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Try to open the booking page directly (Book Facility, FacilityId = 1) without logging in.
Test Input / Data	URL: /Booking/Create?facilityId=1
Expected Result	The request is redirected to the Login page, with a returnUrl pointing back to the booking page so the user returns there after logging in.
Actual Result	Final URL: http://localhost:5233/Account/Login?ReturnUrl=%2FBooking%2FCreate%3FfacilityId%3D1
Test Status	Pass

Evidence	Screenshot: TC06_Guest_Blocked_From_Booking.png
----------	---



Guest redirected to the Login page after attempting to book a facility directly.

TC07 - Guest is blocked from Profile and My Bookings pages

Test Case ID	TC07
Feature / Module	Access Control (Guest)
Test Scenario	Guest is blocked from Profile and My Bookings pages
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Try to open the Profile page (/Profile) directly. 2. Try to open the My Bookings page (/Booking/MyBookings) directly.
Test Input / Data	URLs: /Profile and /Booking/MyBookings
Expected Result	Both pages redirect an unauthenticated visitor to the Login page.
Actual Result	/Profile blocked: true; /Booking/MyBookings blocked: true.
Test Status	Pass
Evidence	No screenshot taken. This check uses the same login-redirect mechanism already shown with a screenshot in TC06, so a second image was not needed.

Note: No screenshot taken. This check uses the same login-redirect mechanism already shown with a screenshot in TC06, so a second image was not needed.

TC08 - Guest submits a contact inquiry successfully

Test Case ID	TC08
Feature / Module	Inquiry (Guest)
Test Scenario	Guest submits a contact inquiry successfully
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Open the Contact Us page. 2. Fill in Name, Email, Subject and Message. 3. Submit the form.
Test Input / Data	Name = "Nadia Perera", Email = "nadia.perera.test@example.com", Subject = "Question about membership fees", Message = "Hello, could you tell me if there is an annual membership fee..."
Expected Result	A confirmation message is shown and the inquiry is saved with Status = "New".
Actual Result	Success confirmation message displayed.
Test Status	Pass
Evidence	Screenshot: TC08_Guest_Inquiry_Success.png


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Thank you — your inquiry has been submitted. We'll get back to you soon.



Contact Us

Have a question about a facility or membership? Send us a message — no account needed.

Your Name

Email

Subject

Message

Send Message


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Confirmation message shown after a guest successfully submits an inquiry.

TC09 - Inquiry form rejects an empty submission

Test Case ID	TC09
Feature / Module	Inquiry (Guest)
Test Scenario	Inquiry form rejects an empty submission
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Open the Contact Us page. 2. Leave all fields empty. 3. Click "Send Message".
Test Input / Data	Name, Email, Subject and Message all left blank.
Expected Result	The form is not submitted. Required-field validation messages are shown for each field.
Actual Result	Required-field validation messages displayed; no record created.
Test Status	Pass
Evidence	Screenshot: TC09_Guest_Inquiry_Validation.png

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Contact Us

Have a question about a facility or membership? Send us a message — no account needed.

The Your Name field is required.
The Email field is required.
The Subject field is required.
The Message field is required.

Your Name

The Your Name field is required.

Email

The Email field is required.

Subject

The Subject field is required.

Message

The Message field is required.

Send Message

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- Contact Us

ACCOUNT

- Login
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Validation messages shown when the inquiry form is submitted empty.

TC10 - Registration is rejected for an email already in use

Test Case ID	TC10
Feature / Module	Registration (Guest)
Test Scenario	Registration is rejected for an email already in use
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Open the Register page. 2. Fill in the form using an email address that already belongs to an existing member (alice.johnson@example.com). 3. Submit the form.
Test Input / Data	Email = "alice.johnson@example.com" (already registered), all other fields valid.
Expected Result	The registration is rejected with the message "This email is already registered." No new account is created.

Actual Result	Error "This email is already registered." displayed.
Test Status	Pass
Evidence	Screenshot: TC10_Registration_Duplicate_Email.png

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Create your account

Register to search by date and time, and start booking facilities.

This email is already registered.

ACCOUNT DETAILS

First Name

Last Name

Email

This email is already registered.

Password

- × At least 8 characters
- × At least one letter
- × At least one digit

Confirm Password

CONTACT DETAILS

Phone

City

Address

PREFERRED SPORTS (optional)

Athletics

Badminton

Basketball

Cricket

Soccer

Swimming

Tennis

Volleyball

Create Account

Already have an account? [Sign in](#)

Registration form showing the duplicate-email error.

TC11 - Registration is rejected for a weak password and mismatched confirmation

Test Case ID	TC11
--------------	------

Feature / Module	Registration (Guest)
Test Scenario	Registration is rejected for a weak password and mismatched confirmation
Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Open the Register page. 2. Fill in the form using Password = "abc" and Confirm Password = "xyz". 3. Submit the form.
Test Input / Data	Password = "abc", Confirm Password = "xyz".
Expected Result	The form is rejected. Validation messages are shown for the password (too short / must contain a letter and a digit) and for the mismatched confirmation.
Actual Result	2 field-level validation message(s) displayed ("Password must contain at least one letter and one digit.", "Passwords do not match.").
Test Status	Pass
Evidence	Screenshot: TC11_Registration_Weak_Password_Mismatch.png

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Create your account

Register to search by date and time, and start booking facilities.

Password must contain at least one letter and one digit.
Passwords do not match.

ACCOUNT DETAILS

First Name

Last Name

Test

Weak

Email

weak.password.test@example.com

Password

At least 8 characters

At least one letter

At least one digit

weak.password.test

Confirm Password

At least 8 characters

At least one letter

At least one digit

weak.password.test

Passwords do not match.

CONTACT DETAILS

Phone

City

+94 77 111 3333

Colombo

Address

2 Test Street

PREFERRED SPORTS (optional)

Athletics

Badminton

Basketball

Cricket

Soccer

Swimming

Tennis

Volleyball

Create Account

Already have an account? [Sign in](#)

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ACCOUNT
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Registration form showing password strength and confirm-password validation errors, with the live password requirement checklist.

TC12 - Guest registers successfully as a new member

Test Case ID	TC12
Feature / Module	Registration (Guest)
Test Scenario	Guest registers successfully as a new member

Preconditions	The application is running. The user has not logged in (Guest).
Test Steps	1. Open the Register page. 2. Fill in valid personal details and a valid password. 3. Select preferred sports (Tennis and Swimming). 4. Submit the form.
Test Input / Data	First Name = "Nadia", Last Name = "Perera", Email = "nadia.perera.test@example.com", Phone = "+94 71 234 5678", Address = "12 Lake Road", City = "Colombo", Password = "Test@1234", Preferred Sports = Tennis, Swimming.
Expected Result	The account is created, the member is automatically signed in, and the browser is redirected to the home page.
Actual Result	Redirected to http://localhost:5233/ after successful registration.
Test Status	Pass
Evidence	Screenshot: TC12_Registration_Success.png

Find and book sports facilities easily.

The Community Sports Council supports local sports programs and manages a network of public facilities available to every resident.

 Find Facilities

 Browse with Availability

Popular Sports



Athletics



Badminton



Basketball



Cricket



Soccer



Swimming



Tennis

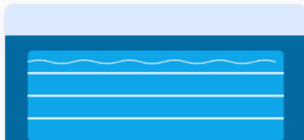


Volleyball

Featured Facilities

7 active facilities currently available to book.

[View all facilities →](#)



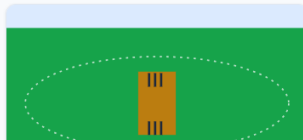
Central Community Pool

 City Centre, Springfield

Swimming Pool

Swimming

[Details](#)



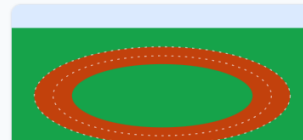
Eastfield Cricket Ground

 Eastfield, Springfield

Cricket Ground

Cricket

[Details](#)



Northgate Athletics Track

 Northgate, Springfield

Athletics Track

Athletics Cricket

[Details](#)

How It Works



Search

Browse or search facilities by sport and location.



Choose a facility

View details, sports supported, and full facility information.



Select a time

Check real-time availability for the date and time you want.



Confirm

Confirm your booking instantly — no waiting for approval.

Why Book With Us



Easy booking

A simple, guided flow from search to confirmation.



Real-time availability

See exactly which time slots are free before you book.



Multiple sports

8 sports across our facility network.



Secure member accounts

Your bookings and profile are private to you.


Community Sports Facilities Booking System
 Find, explore and book local sports facilities in a few clicks.

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Home page shown immediately after a successful registration, with the new member signed in.

15.2 Member Module

TC13 - Registered member logs in with correct credentials

Test Case ID	TC13
Feature / Module	Login (Member)
Test Scenario	Registered member logs in with correct credentials
Preconditions	The application is running. The member account from TC12 already exists.
Test Steps	1. Open the Login page. 2. Enter the correct email and password. 3. Submit the form.
Test Input / Data	Email = "nadia.perera.test@example.com", Password = "Test@1234".
Expected Result	Login succeeds. The user is redirected to the home page and the navigation bar shows the member's name.
Actual Result	Redirected to http://localhost:5233/; nav shows signed-in member name.
Test Status	Pass
Evidence	Screenshot: TC13_Member_Login_Success.png

Find and book sports facilities easily.

The Community Sports Council supports local sports programs and manages a network of public facilities available to every resident.

Find Facilities

Browse with Availability

Popular Sports



Athletics



Badminton



Basketball



Cricket



Soccer



Swimming



Tennis

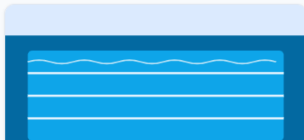


Volleyball

Featured Facilities

7 active facilities currently available to book.

[View all facilities →](#)



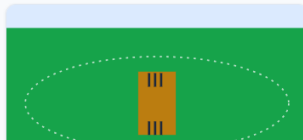
Central Community Pool

City Centre, Springfield

Swimming Pool

Swimming

[Details](#)



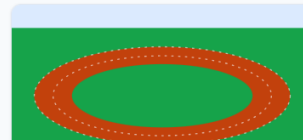
Eastfield Cricket Ground

Eastfield, Springfield

Cricket Ground

Cricket

[Details](#)



Northgate Athletics Track

Northgate, Springfield

Athletics Track

Athletics Cricket

[Details](#)

How It Works



Search

Browse or search facilities by sport and location.



Choose a facility

View details, sports supported, and full facility information.



Select a time

Check real-time availability for the date and time you want.



Confirm

Confirm your booking instantly — no waiting for approval.

Why Book With Us



Easy booking

A simple, guided flow from search to confirmation.



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See exactly which time slots are free before you book.



Multiple sports

8 sports across our facility network.



Secure member accounts

Your bookings and profile are private to you.

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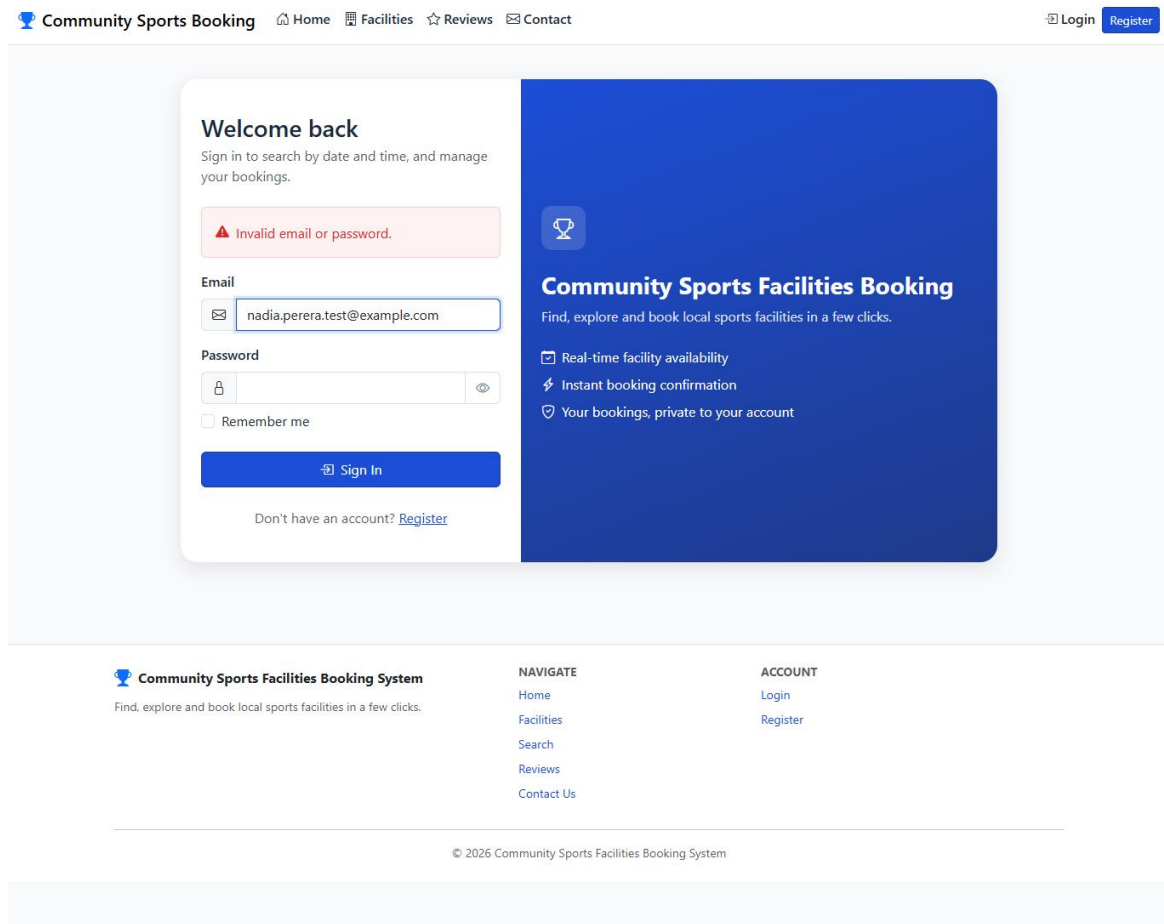
ACCOUNT

[My Bookings](#)
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Home page after a successful member login, with the member's name shown in the navigation bar.

TC14 - Login is rejected for a valid email with the wrong password

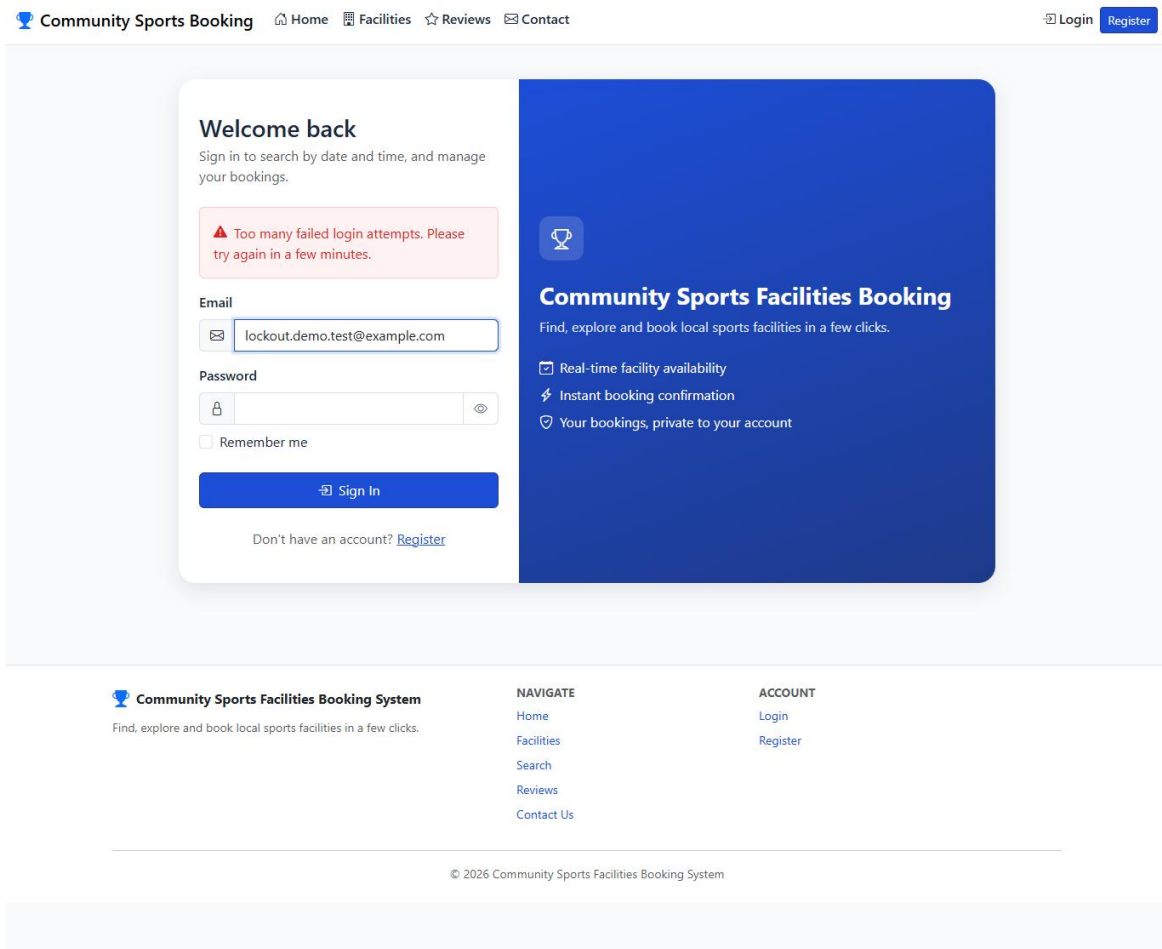
Test Case ID	TC14
Feature / Module	Login (Member)
Test Scenario	Login is rejected for a valid email with the wrong password
Preconditions	The application is running. The member account from TC12 already exists.
Test Steps	1. Open the Login page. 2. Enter the correct email but an incorrect password. 3. Submit the form.
Test Input / Data	Email = "nadia.perera.test@example.com", Password = "WrongPassword1".
Expected Result	Login is rejected with a generic message ("Invalid email or password.") that does not reveal whether the email or the password was wrong.
Actual Result	Generic "Invalid email or password." message shown.
Test Status	Pass
Evidence	Screenshot: TC14_Member_Login_Wrong_Password.png



Login page showing the generic invalid-credentials error.

TC15 - Account is temporarily locked out after repeated failed logins

Test Case ID	TC15
Feature / Module	Login (Member)
Test Scenario	Account is temporarily locked out after repeated failed logins
Preconditions	The application is running. A non-existent email address is used, to show that lockout applies equally whether or not the email belongs to a real account.
Test Steps	1. Submit the login form with the wrong password 6 times in a row, using the same email address.
Test Input / Data	Email = "lockout.demo.test@example.com", Password = "WrongPassword1" (submitted 6 times).
Expected Result	After 5 failed attempts the account is temporarily locked. The 6th attempt shows "Too many failed login attempts. Please try again in a few minutes." instead of the normal invalid-credentials message.
Actual Result	Lockout message shown starting on the 6th attempt (the 5th failed attempt reaches the internal threshold but still shows the generic error; the following attempt is blocked before credentials are even checked).
Test Status	Pass
Evidence	Screenshot: TC15_Member_Login_Lockout.png



Login page showing the temporary lockout message after repeated failed attempts.

TC16 - Member views their own profile

Test Case ID	TC16
Feature / Module	Profile (Member)
Test Scenario	Member views their own profile
Preconditions	The application is running. The user is logged in as a Member.
Test Steps	1. Open the Profile page.
Test Input / Data	None.
Expected Result	The page shows the member's personal details, their email address as read-only, and their preferred sports.
Actual Result	Profile shows correct email (read-only) and preferred sports.
Test Status	Pass
Evidence	Screenshot: TC16_Member_Profile_View.png

Community Sports Booking
Home
Facilities
Reviews
Contact
NP Nadia Perera

NP
Nadia Perera
Member since September 2026

PERSONAL INFORMATION

Email
nadia.perera.test@example.com
Email cannot be changed.

FirstName
Nadia

LastName
Perera

Phone
+94 71 234 5678

City
Colombo

AddressLine
12 Lake Road

Save Changes

PREFERRED SPORTS

Athletics
Badminton
Basketball
Cricket

Soccer
Swimming
Tennis
Volleyball

Update Preferences

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My Bookings
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Member profile page showing personal details and preferred sports.

TC17 - Member updates their personal information

Test Case ID	TC17
Feature / Module	Profile (Member)
Test Scenario	Member updates their personal information
Preconditions	The application is running. The user is logged in as a Member.
Test Steps	1. Open the Profile page. 2. Change the phone number and address. 3. Click "Save Changes".
Test Input / Data	Phone = "+94 71 999 8888", Address = "45 Galle Road", City = "Colombo".
Expected Result	The updated details are saved and shown back on the profile page. The email address cannot be changed.

Actual Result	Updated address and phone number now shown on profile.
Test Status	Pass
Evidence	Screenshot: TC17_Member_Profile_Edit_Success.png

Community Sports Booking

HomeFacilitiesReviewsContact

NPNadia Perera

NP

Nadia Perera

Member since September 2026

PERSONAL INFORMATION

Email

nadia.perera.test@example.com

Email cannot be changed.

FirstName

Nadia

LastName

Perera

Phone

+94 71 999 8888

City

Colombo

AddressLine

45 Galle Road

Save Changes

PREFERRED SPORTS

Athletics

Badminton

Basketball

Cricket

Soccer

Swimming

Tennis

Volleyball

Update Preferences

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Contact Us

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My Bookings

Profile

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Profile page showing the updated address and phone number after saving.

TC18 - Profile update is rejected for an invalid phone number

Test Case ID	TC18
Feature / Module	Profile (Member)
Test Scenario	Profile update is rejected for an invalid phone number
Preconditions	The application is running. The user is logged in as a Member.
Test Steps	1. Open the Profile page. 2. Enter an invalid phone number containing letters and symbols. 3. Click "Save Changes".

Test Input / Data	Phone = "not-a-phone-###".
Expected Result	The update is rejected with a validation message explaining that the phone number may only contain digits, spaces, +, - and parentheses. The invalid value is not saved.
Actual Result	2 field validation message(s) displayed; invalid value not saved.
Test Status	Pass
Evidence	Screenshot: TC18_Member_Profile_Validation.png

Community Sports Booking
Home
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Reviews
Contact
NP Nadia Perera

NP Nadia Perera
Member since September 2026

PERSONAL INFORMATION

Phone may only contain digits, spaces, +, -, and parentheses.

Email
nadia.perera.test@example.com
Email cannot be changed.

FirstName
Nadia
LastName
Perera

Phone
not-a-phone-###
City
Colombo
Phone may only contain digits, spaces, +, -, and parentheses.

AddressLine
45 Galle Road

Save Changes

PREFERRED SPORTS

Athletics
Badminton
Basketball
Cricket
Soccer
Swimming
Tennis
Volleyball

Update Preferences

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ACCOUNT
My Bookings
Profile

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Profile page showing the phone number validation error.

TC19 - Member adds an additional preferred sport

Test Case ID	TC19
--------------	------

Feature / Module	Sports Preferences (Member)
Test Scenario	Member adds an additional preferred sport
Preconditions	The application is running. The user is logged in as a Member.
Test Steps	1. Open the Profile page. 2. Select "Basketball" in the Preferred Sports section, in addition to the sports already selected. 3. Click "Update Preferences".
Test Input / Data	Selected sports = Tennis, Swimming, Basketball.
Expected Result	The preference list is saved. Basketball is now shown as selected alongside the previous sports.
Actual Result	Basketball now shown among preferred sports.
Test Status	Pass
Evidence	Screenshot: TC19_Member_Sports_Preferences_Updated.png

 Community Sports Booking
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[Reviews](#)
[Contact](#)

 Nadia Perera


Nadia Perera
 Member since September 2026

 PERSONAL INFORMATION

Email

 nadia.perera.test@example.com

Email cannot be changed.

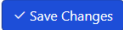
FirstName

LastName

Phone

City

AddressLine



 PREFERRED SPORTS

☐ Athletics
 ☐ Badminton
 ☒ Basketball
 ☐ Cricket

☐ Soccer
 ☒ Swimming
 ☒ Tennis
 ☐ Volleyball



 Community Sports Facilities Booking System
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Profile page showing Basketball added to the member's preferred sports.

TC20 - Signed-in member searches with a specific date and time window

Test Case ID	TC20
Feature / Module	Facility Search (Member)
Test Scenario	Signed-in member searches with a specific date and time window
Preconditions	The application is running. The user is logged in as a Member.
Test Steps	1. Open the Search Facilities page. 2. Confirm that date and time fields are shown (unlike the guest search). 3. Enter "Tennis" and a future date/time window. 4. Submit the search.
Test Input / Data	Facility type = "Tennis", Date = 16 Sep 2026, Start = 09:00, End = 10:00.
Expected Result	The date and time fields are available to a signed-in member. The matching facility is returned, annotated with real-time availability for the requested window.
Actual Result	Date/Time fields present for a member; matching facility returned with availability annotation.
Test Status	Pass
Evidence	Screenshot: TC20_Member_Search_With_Availability.png

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Nadia Perera

Search Facilities

Search by type or location, or add a date and time to see real-time availability.

Sport or Facility Type

Tennis

Location

e.g. Springfield

Date

09/16/2026

Start Time

09:00 AM

End Time

10:00 AM

Search

Clear

1 facility found.

Riverside Tennis Courts

Available

Riverside, Springfield

Tennis Court

Tennis

Details

Book

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Member search results showing availability for the requested date and time.

TC21 - Member books an available facility

Test Case ID	TC21
Feature / Module	Booking (Member)
Test Scenario	Member books an available facility
Preconditions	The application is running. The user is logged in as a Member.
Test Steps	1. Open the booking page for "Riverside Tennis Courts". 2. Enter a future date and time. 3. Click "Confirm Booking".
Test Input / Data	Facility = Riverside Tennis Courts, Date = 16 Sep 2026, Start = 09:00, End = 10:00.
Expected Result	The booking is created successfully. The user is redirected to My Bookings, where the new booking appears under "Upcoming".
Actual Result	Redirected to My Bookings; new booking visible under Upcoming.
Test Status	Pass

Evidence	Screenshot: TC21_Member_Booking_Created.png
----------	---

Community Sports Booking Home Facilities Reviews Contact **NP Nadia Perera**

My Bookings

Upcoming

- Riverside Tennis Courts**
Tennis Court · Riverside
16 Sep 2026 09:00–10:00 **Upcoming** [Cancel](#)

Completed

- Central Community Pool**
Swimming Pool · City Centre
20 Aug 2026 10:00–11:00 [Leave a Review](#)
- Oakwood Sports Hall**
Sports Hall · Oakwood
25 Aug 2026 14:00–15:00 [Leave a Review](#)

Cancelled

You have no cancelled bookings.

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My Bookings page showing the newly created upcoming booking.

TC22 - Booking form rejects a past date

Test Case ID	TC22
Feature / Module	Booking Validation (Member)
Test Scenario	Booking form rejects a past date
Preconditions	The application is running. The user is logged in as a Member.
Test Steps	1. Open the booking page for "Riverside Tennis Courts". 2. Try to enter a date that has already passed. 3. Click "Confirm Booking".
Test Input / Data	Date = 01 Sep 2026 (in the past relative to the test date).
Expected Result	The browser itself blocks the submission, since the date field will not accept a value earlier than today. This front-end rule mirrors the same rule enforced on the server.
Actual Result	Browser displayed a native "value must be on or after 2026-09-08" validation message; the form was not submitted.

Test Status	Pass
Evidence	Screenshot: TC22_Booking_Past_Date_Rejected.png

Community Sports Booking Home Facilities Reviews Contact NP Nadia Perera

← Back to facility

Book: Riverside Tennis Courts

Choose a date and time window for this facility.

⚠ Please enter a value greater than or equal to 2026-09-08.

Date: 09/01/2026

Please enter a value greater than or equal to 2026-09-08.

Quick time picker (optional)

08:00-09:00 09:00-10:00 12:00-13:00 17:00-18:00 18:00-19:00 19:00-20:00

Start Time: 09:00 AM End Time: 10:00 AM

Confirm Booking Cancel

Riverside Tennis Courts
Tennis Court

Date: 2026-09-01
Time: 09:00 – 10:00

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My Bookings
Profile

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Booking form showing the browser's built-in validation message for a past date.

TC23 - Booking is rejected when it overlaps an existing booking

Test Case ID	TC23
Feature / Module	Booking Validation (Member)
Test Scenario	Booking is rejected when it overlaps an existing booking
Preconditions	The application is running. The user is logged in as a Member. A booking already exists for Riverside Tennis Courts on 16 Sep 2026, 09:00-10:00 (created in TC21).
Test Steps	1. Open the booking page for "Riverside Tennis Courts". 2. Enter the same date and time as the existing booking. 3. Click "Confirm Booking".
Test Input / Data	Facility = Riverside Tennis Courts, Date = 16 Sep 2026, Start = 09:00, End = 10:00 (same slot as TC21).
Expected Result	The booking is rejected with the message that the facility is no longer available for the selected time.

Actual Result	Error "This facility is no longer available for the selected time — please choose another slot." displayed.
Test Status	Pass
Evidence	Screenshot: TC23_Booking_Overlap_Rejected.png

Community Sports Booking | Home | Facilities | Reviews | Contact | NP Nadia Perera

← Back to facility

Book: Riverside Tennis Courts

Choose a date and time window for this facility.

⚠ This facility is no longer available for the selected time — please choose another slot.

Date
09/16/2026

Quick time picker (optional)
 08:00-09:00 | 09:00-10:00 | 12:00-13:00 | 17:00-18:00 | 18:00-19:00 | 19:00-20:00

Start Time 09:00 AM **End Time** 10:00 AM

[Confirm Booking](#) [Cancel](#)

Riverside Tennis Courts
Tennis Court

Date: 2026-09-16
Time: 09:00 – 10:00

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Booking form showing the overlap-rejection error for a slot that is already booked.

TC24 - Member views their bookings grouped by status

Test Case ID	TC24
Feature / Module	My Bookings (Member)
Test Scenario	Member views their bookings grouped by status
Preconditions	The application is running. The user is logged in as a Member. The member has more than one upcoming booking.
Test Steps	1. Open the My Bookings page.
Test Input / Data	None.
Expected Result	Bookings are grouped into Upcoming, Completed and Cancelled sections, with each booking showing the correct facility, date and time.
Actual Result	Both upcoming bookings listed under the Upcoming section.
Test Status	Pass

Evidence	Screenshot: TC24_Member_MyBookings.png
----------	--

Community Sports Booking Home Facilities Reviews Contact **NP Nadia Perera**

My Bookings

Upcoming

- Riverside Tennis Courts**
Tennis Court · Riverside
16 Sep 2026 09:00–10:00 **Upcoming** [Cancel](#)
- Northgate Athletics Track**
Athletics Track · Northgate
18 Sep 2026 14:00–15:00 **Upcoming** [Cancel](#)

Completed

- Central Community Pool**
Swimming Pool · City Centre
20 Aug 2026 10:00–11:00 [Leave a Review](#)
- Oakwood Sports Hall**
Sports Hall · Oakwood
25 Aug 2026 14:00–15:00 [Leave a Review](#)

Cancelled

You have no cancelled bookings.

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My Bookings page showing bookings grouped under Upcoming, Completed and Cancelled.

TC25 - Member cancels their own upcoming booking

Test Case ID	TC25
Feature / Module	Booking Cancellation (Member)
Test Scenario	Member cancels their own upcoming booking
Preconditions	The application is running. The user is logged in as a Member.
Test Steps	1. Open the My Bookings page. 2. Click "Cancel" on the Riverside Tennis Courts booking. 3. Confirm the cancellation.
Test Input / Data	Booking = Riverside Tennis Courts, 16 Sep 2026, 09:00-10:00.
Expected Result	A confirmation message is shown and the booking moves from Upcoming to the Cancelled section, instead of being deleted.
Actual Result	Success message shown and booking moved to Cancelled section.

Test Status	Pass
Evidence	Screenshot: TC25_Member_Booking_Cancelled.png

[Home](#)
[Facilities](#)
[Reviews](#)
[Contact](#)

NP Nadia Perera

Your booking has been cancelled.

My Bookings

Upcoming

Northgate Athletics Track
Athletics Track · Northgate

18 Sep 2026
14:00–15:00

Upcoming

Cancel

Completed

Central Community Pool
Swimming Pool · City Centre

20 Aug 2026
10:00–11:00

Leave a Review

Oakwood Sports Hall
Sports Hall · Oakwood

25 Aug 2026
14:00–15:00

Leave a Review

Cancelled

Riverside Tennis Courts
Tennis Court · Riverside

16 Sep 2026
09:00–10:00

Cancelled

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My Bookings page showing the booking moved to the Cancelled section after cancellation.

TC26 - Member submits a review for a completed booking

Test Case ID	TC26
Feature / Module	Review Submission (Member)
Test Scenario	Member submits a review for a completed booking
Preconditions	The application is running. The user is logged in as a Member. A completed booking exists for Central Community Pool, 20 Aug 2026, with no review yet.
Test Steps	1. Open the My Bookings page. 2. Click "Leave a Review" on the completed Central Community Pool booking. 3. Select a rating and enter a comment. 4. Click "Submit Review".

Test Input / Data	Rating = 4 stars, Comment = "Clean and well-maintained pool with friendly staff. Would visit again."
Expected Result	The review is saved. A confirmation message is shown and the booking is now marked "Reviewed".
Actual Result	Success message shown; booking now marked Reviewed.
Test Status	Pass
Evidence	Screenshot: TC26_Member_Review_Submitted.png

[Home](#)
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[Contact](#)

Nadia Perera

Thank you — your review has been submitted.

My Bookings

Upcoming

Northgate Athletics Track
Athletics Track · Northgate

18 Sep 2026
14:00–15:00

Upcoming
Cancel

Completed

Central Community Pool
Swimming Pool · City Centre

20 Aug 2026
10:00–11:00

Reviewed

Oakwood Sports Hall
Sports Hall · Oakwood

25 Aug 2026
14:00–15:00

Leave a Review

Cancelled

Riverside Tennis Courts
Tennis Court · Riverside

16 Sep 2026
09:00–10:00

Cancelled

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My Bookings page showing the confirmation message and the booking marked as Reviewed.

TC27 - Review submission is rejected when rating and comment are missing

Test Case ID	TC27
Feature / Module	Review Validation (Member)
Test Scenario	Review submission is rejected when rating and comment are missing
Preconditions	The application is running. The user is logged in as a Member. A second completed, unreviewed booking exists for Oakwood Sports Hall.
Test Steps	1. Open the review page for the Oakwood Sports Hall booking.

	2. Leave the rating and comment empty. 3. Click "Submit Review".
Test Input / Data	Rating = (none selected), Comment = (empty).
Expected Result	The review is rejected. Validation messages are shown for the missing rating and the missing comment.
Actual Result	2 field validation message(s) displayed ("Please select a rating.", "Please enter a comment."); no review saved.
Test Status	Pass
Evidence	Screenshot: TC27_Review_Validation_Missing_Fields.png

Community Sports Booking Home Facilities **Reviews** Contact

NP Nadia Perera

Leave a Review

Share your experience at **Oakwood Sports Hall**.

Facility
Oakwood Sports Hall

Booking Date
25 Aug 2026 · 14:00–15:00

Please enter a comment.

Rating

★ ★ ★ ★ ★

Comment

What did you think of this facility?

Please enter a comment.

Submit Review Cancel

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Review form showing validation messages for a missing rating and comment.

TC28 - Signed-in member logs out

Test Case ID	TC28
Feature / Module	Logout (Member)
Test Scenario	Signed-in member logs out
Preconditions	The application is running. The user is logged in as a Member.

Test Steps	1. Click the member's name in the navigation bar. 2. Click "Logout".
Test Input / Data	None.
Expected Result	The session ends and the navigation bar reverts to showing "Login" and "Register" links.
Actual Result	Nav shows Login/Register; member-only session ended.
Test Status	Pass
Evidence	Screenshot: TC28_Member_Logout.png

Find and book sports facilities easily.

The Community Sports Council supports local sports programs and manages a network of public facilities available to every resident.

[Find Facilities](#)

[Browse with Availability](#)

Popular Sports



Athletics



Badminton



Basketball



Cricket



Soccer



Swimming



Tennis

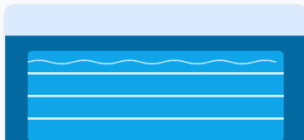


Volleyball

Featured Facilities

7 active facilities currently available to book.

[View all facilities →](#)



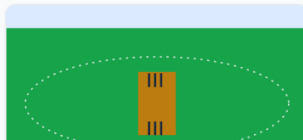
Central Community Pool

City Centre, Springfield

Swimming Pool

Swimming

[Details](#)



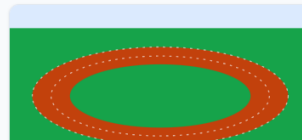
Eastfield Cricket Ground

Eastfield, Springfield

Cricket Ground

Cricket

[Details](#)



Northgate Athletics Track

Northgate, Springfield

Athletics Track

Athletics Cricket

[Details](#)

How It Works



Search

Browse or search facilities by sport and location.



Choose a facility

View details, sports supported, and full facility information.



Select a time

Check real-time availability for the date and time you want.



Confirm

Confirm your booking instantly — no waiting for approval.

Why Book With Us



Easy booking

A simple, guided flow from search to confirmation.



Real-time availability

See exactly which time slots are free before you book.



Multiple sports

8 sports across our facility network.



Secure member accounts

Your bookings and profile are private to you.

[Become a Member](#)

Register to search facilities by date and time, and make bookings.

[Register](#)

NAVIGATE

[Home](#)
[Facilities](#)
[Search](#)
[Reviews](#)
[Contact Us](#)

ACCOUNT

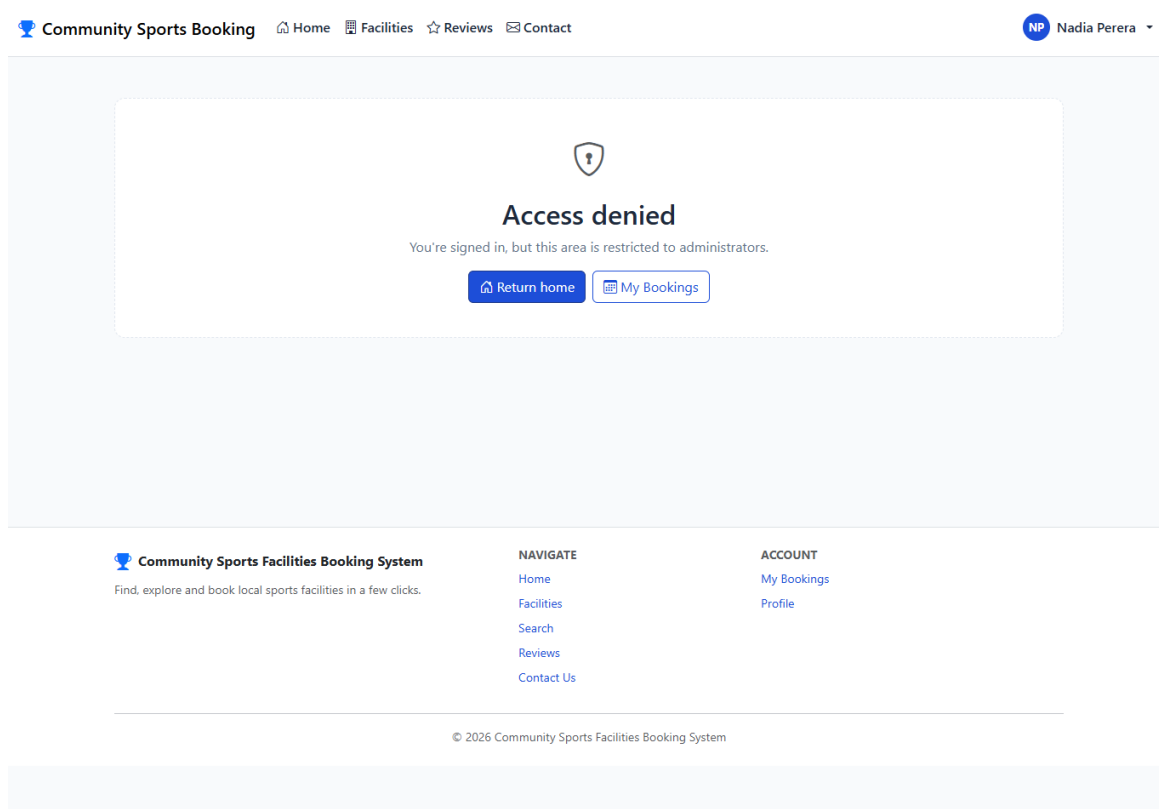
[Login](#)
[Register](#)

Home page navigation bar after logout, showing Login and Register links again.

15.3 Administrator Module

TC29 - A signed-in, non-admin member is denied access to the Admin area

Test Case ID	TC29
Feature / Module	Access Control (Admin)
Test Scenario	A signed-in, non-admin member is denied access to the Admin area
Preconditions	The application is running. A member without administrator rights is logged in.
Test Steps	1. While logged in as an ordinary member, try to open the Admin area (/Admin) directly.
Test Input / Data	URL: /Admin
Expected Result	The member is redirected to an "Access denied" page rather than being shown the admin dashboard.
Actual Result	Redirected to Access Denied page.
Test Status	Pass
Evidence	Screenshot: TC29_NonAdmin_Access_Denied.png



Access denied page shown to a signed-in member without administrator rights.

TC30 - The dedicated administrator account logs in and is sent to the Dashboard

Test Case ID	TC30
Feature / Module	Login (Admin)
Test Scenario	The dedicated administrator account logs in and is sent to the Dashboard
Preconditions	The application is running. The user is not logged in.
Test Steps	1. Open the Login page. 2. Enter the administrator credentials. 3. Submit the form.
Test Input / Data	Email = "admin", Password = "admin".
Expected Result	Login succeeds and the browser is redirected straight to the Admin Dashboard rather than the public home page.
Actual Result	Redirected to http://localhost:5233/Admin.
Test Status	Pass
Evidence	Screenshot: TC30_Admin_Login_Redirect.png

Community Sports Booking **Admin** [View site](#) **SA** System Administrator ▾

Dashboard

A live snapshot of the system — every number below is read from the database on each request.

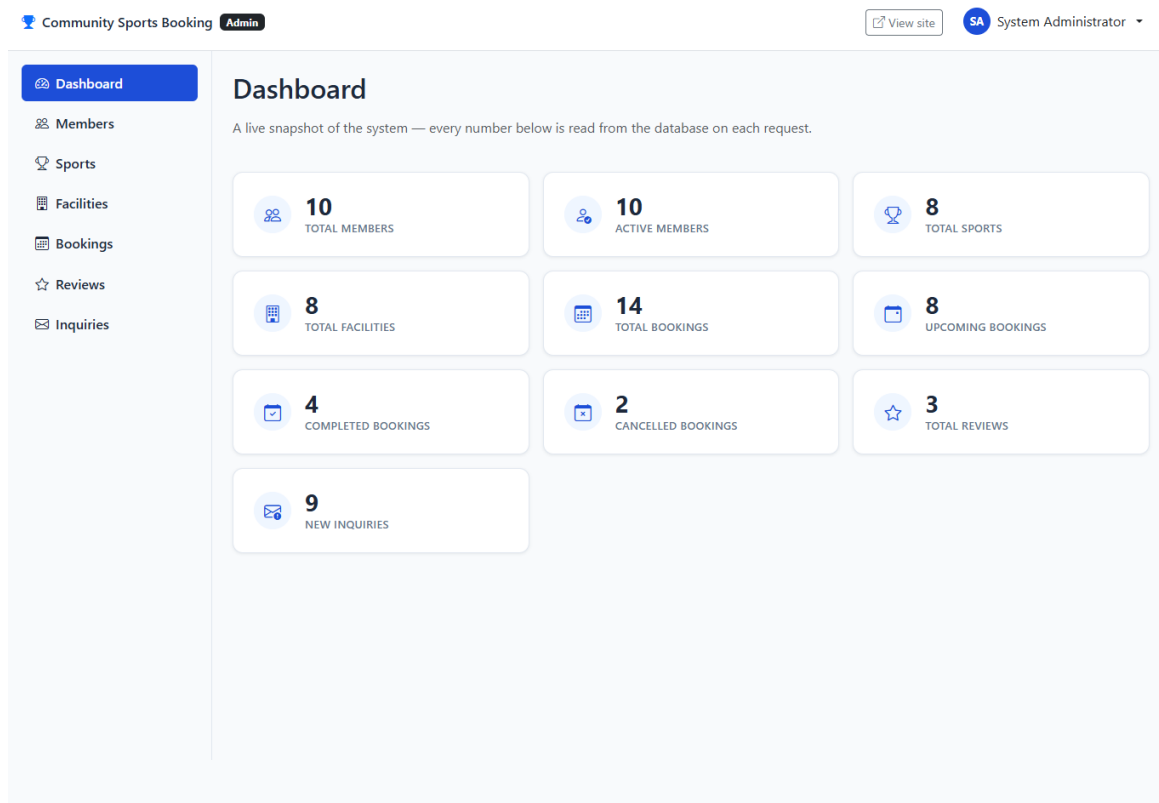
10 TOTAL MEMBERS	10 ACTIVE MEMBERS	8 TOTAL SPORTS
8 TOTAL FACILITIES	14 TOTAL BOOKINGS	8 UPCOMING BOOKINGS
4 COMPLETED BOOKINGS	2 CANCELLED BOOKINGS	3 TOTAL REVIEWS
9 NEW INQUIRIES		

Admin Dashboard shown immediately after the administrator logs in.

TC31 - Admin Dashboard displays live system statistics

Test Case ID	TC31
Feature / Module	Dashboard (Admin)
Test Scenario	Admin Dashboard displays live system statistics

Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).
Test Steps	1. Open the Admin Dashboard.
Test Input / Data	None.
Expected Result	The dashboard shows live counts read from the database, including total and active members, total sports, total and active facilities, total/upcoming/completed/cancelled bookings, total reviews, and new inquiries.
Actual Result	Dashboard rendered member/booking/facility/review/inquiry statistic cards with real counts.
Test Status	Pass
Evidence	Screenshot: TC31_Admin_Dashboard.png



Admin Dashboard showing live statistic cards for members, facilities, bookings, reviews and inquiries.

TC32 - Admin searches for a member and views their details

Test Case ID	TC32
Feature / Module	Member Management (Admin)
Test Scenario	Admin searches for a member and views their details
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).

Test Steps	1. Open Admin > Members. 2. Search for "Nadia". 3. Open the matching member's details page.
Test Input / Data	Search text = "Nadia".
Expected Result	The search returns the matching member. The details page shows their profile information, bookings and reviews.
Actual Result	Search returned Nadia Perera; details page shows profile, bookings and reviews.
Test Status	Pass
Evidence	Screenshot: TC32_Admin_Member_Search_Details.png

The screenshot displays the 'Admin' interface for 'Community Sports Booking'. The user is logged in as 'System Administrator'. The left sidebar contains navigation links: Dashboard, Members (selected), Sports, Facilities, Bookings, Reviews, and Inquiries. The main content area shows the details for member 'Nadia Perera', who is 'Active'. A 'Deactivate' button is visible. The profile section includes fields for Email, Phone, Address, and Registered date, along with 'Preferred Sports' (Basketball, Swimming, Tennis). Below this is a 'Bookings (4)' section listing four bookings with their dates, times, and statuses (Upcoming or Completed). At the bottom is a 'Reviews (1)' section showing a review for the 'Central Community Pool'.

Admin member details page showing profile, booking history and reviews for the searched member.

TC33 - Admin deactivates a member account

Test Case ID	TC33
Feature / Module	Member Management (Admin)
Test Scenario	Admin deactivates a member account
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).

Test Steps	1. Open the member's details page. 2. Click "Deactivate". 3. Confirm the action.
Test Input / Data	Member = Nadia Perera.
Expected Result	The member is marked Inactive and a confirmation message is shown.
Actual Result	Success message shown; member now shows an Inactive badge.
Test Status	Pass
Evidence	Screenshot: TC33_Admin_Deactivates_Member.png

Community Sports Booking
Admin
View site
SA System Administrator

Dashboard
Members
Sports
Facilities
Bookings
Reviews
Inquiries

Nadia Perera has been deactivated.

Back to Members

Nadia Perera Inactive
Activate

Email: nadia.perera.test@example.com
Phone: +94 71 999 8888
Address: 45 Galle Road, Colombo
Registered: 08 Sep 2026
Preferred Sports: Basketball, Swimming, Tennis

Bookings (4)

Northgate Athletics Track
Athletics Track
18 Sep 2026
14:00–15:00
Upcoming

Riverside Tennis Courts
Tennis Court
16 Sep 2026
09:00–10:00
Upcoming

Oakwood Sports Hall
Sports Hall
25 Aug 2026
14:00–15:00
Completed

Central Community Pool
Swimming Pool
20 Aug 2026
10:00–11:00
Completed

Reviews (1)

Central Community Pool · Swimming Pool
08 Sep 2026
★★★★☆
Clean and well-maintained pool with friendly staff. Would visit again.

Member details page showing the Inactive badge and confirmation message after deactivation.

TC34 - Admin cannot deactivate their own account

Test Case ID	TC34
Feature / Module	Member Management (Admin)
Test Scenario	Admin cannot deactivate their own account
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).

Test Steps	1. Open the administrator's own member details page. 2. Observe that the deactivate control is disabled ("This is your account"). 3. Send the deactivation request directly to the server (bypassing the disabled button) to confirm the rule is also enforced server-side.
Test Input / Data	Member = System Administrator (the currently signed-in account).
Expected Result	The account cannot be deactivated by any route. The account remains Active and, if attempted directly, the server responds with "You cannot deactivate your own account."
Actual Result	Admin account remains Active after the direct request; server message "You cannot deactivate your own account." was shown.
Test Status	Pass
Evidence	Screenshot: TC34a_Admin_Self_Deactivate_Button_Disabled.png, TC34b_Admin_Self_Deactivate_Blocked.png

Community Sports Booking
Admin
View site
SA System Administrator

Dashboard
Members
Sports
Facilities
Bookings
Reviews
Inquiries

Back to Members
System Administrator
Admin
Active
This is your account

Email admin	Phone 0000000000	Address N/A, N/A	Registered 08 Sep 2026
----------------	---------------------	---------------------	---------------------------

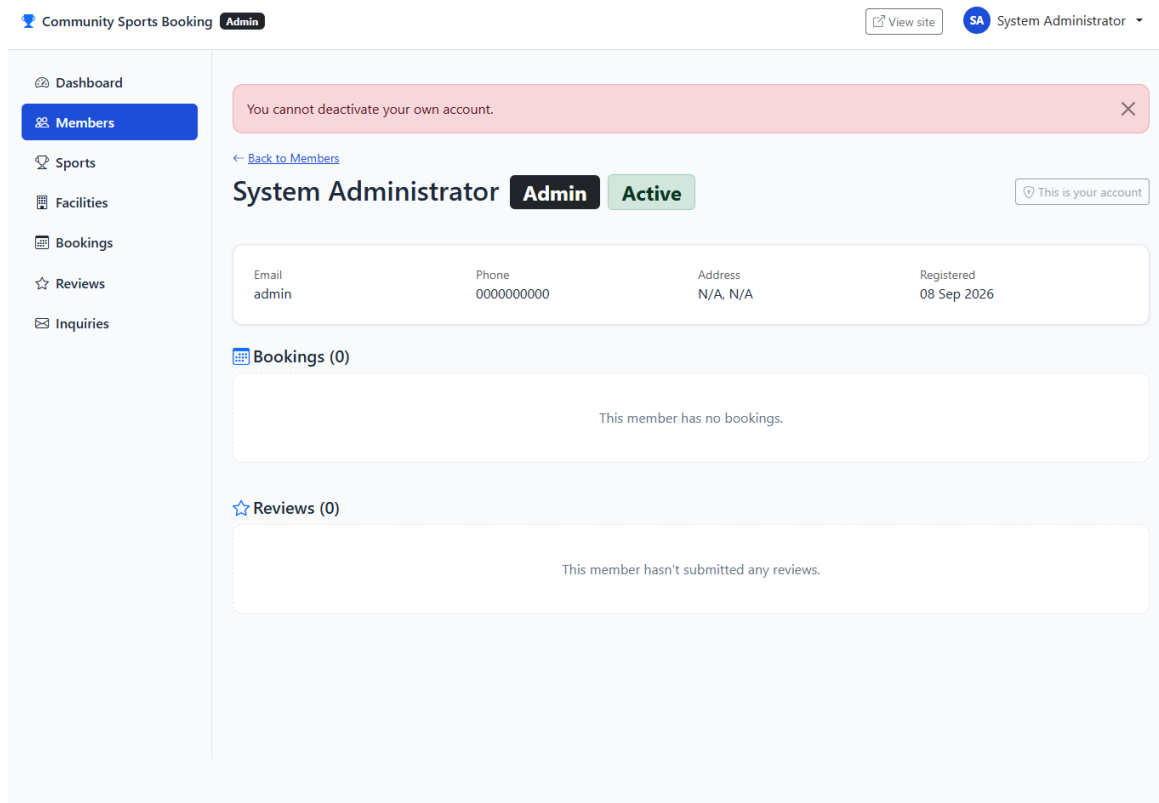
Bookings (0)

This member has no bookings.

Reviews (0)

This member hasn't submitted any reviews.

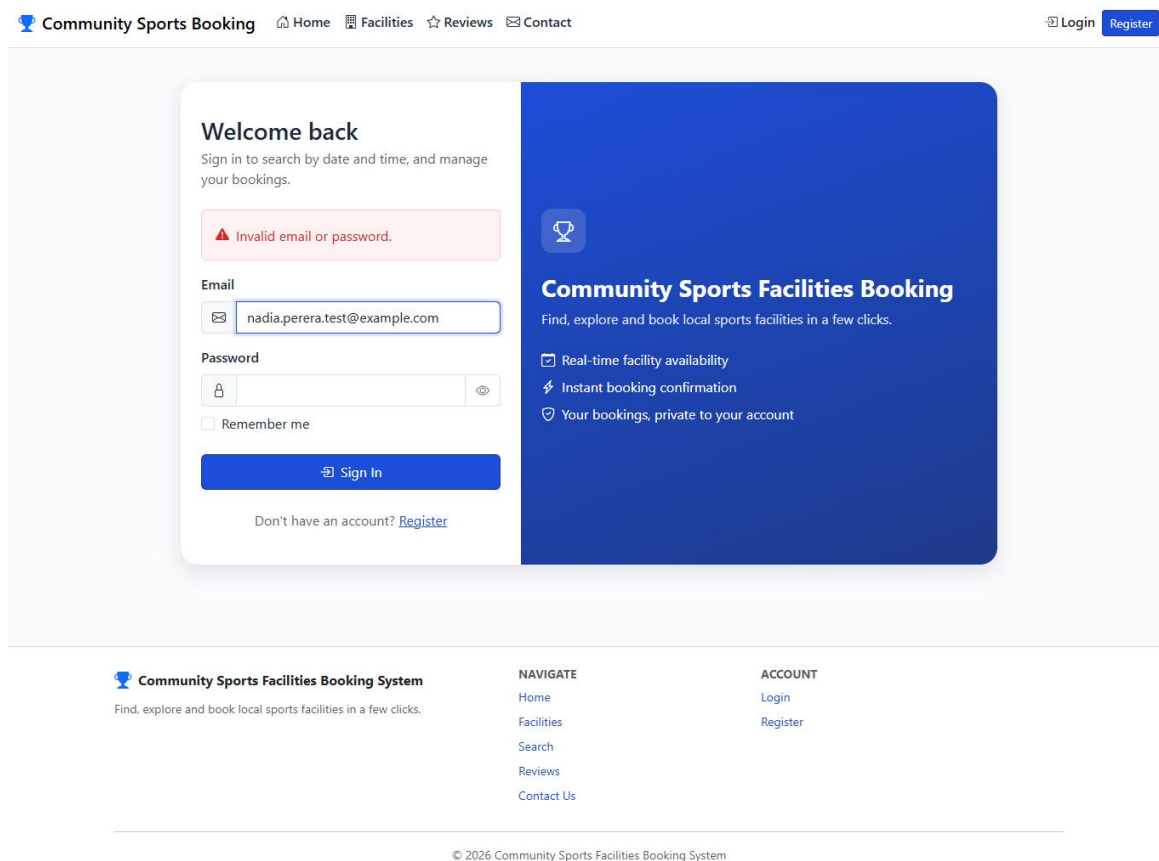
Administrator details page showing the "This is your account" disabled control instead of a Deactivate button.



Same page after a direct request to the endpoint, confirming the account remains Active and showing the self-deactivation error message.

TC35 - A deactivated member can no longer log in

Test Case ID	TC35
Feature / Module	Login (Member)
Test Scenario	A deactivated member can no longer log in
Preconditions	The member account was deactivated by an administrator in TC33.
Test Steps	1. Open the Login page. 2. Enter the deactivated member's correct email and password. 3. Submit the form.
Test Input / Data	Email = "nadia.perera.test@example.com", Password = "Test@1234".
Expected Result	Login is rejected using the same generic message as a wrong password, so a deactivated account cannot be distinguished from a wrong password by a visitor.
Actual Result	Login rejected with the standard generic "Invalid email or password." message.
Test Status	Pass
Evidence	Screenshot: TC35_Deactivated_Member_Login_Blocked.png



Login page showing that a deactivated member is rejected with the same generic error as any other failed login.

TC36 - Admin adds a new sport

Test Case ID	TC36
Feature / Module	Sport Management (Admin)
Test Scenario	Admin adds a new sport
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).
Test Steps	1. Open Admin > Sports > Add Sport. 2. Enter a new sport name. 3. Click "Add Sport".
Test Input / Data	Sport Name = "Table Tennis".
Expected Result	The sport is added to the catalogue and a confirmation message is shown.
Actual Result	Success message shown; "Table Tennis" now listed.
Test Status	Pass
Evidence	Screenshot: TC36_Admin_Sport_Created.png

Community Sports Booking
Admin
View site
SA System Administrator

Dashboard
Members
Sports
Facilities
Bookings
Reviews
Inquiries

Table Tennis has been added.

Sports
9 sports total.
Add Sport

Search by name
e.g. Tennis
Search
Clear

Sport	Facilities	Member Preferences	
Athletics	1	1	View
Badminton	2	1	View
Basketball	1	2	View
Cricket	2	1	View
Soccer	1	1	View
Swimming	1	3	View
Table Tennis	0	0	View
Tennis	1	3	View
Volleyball	2	1	View

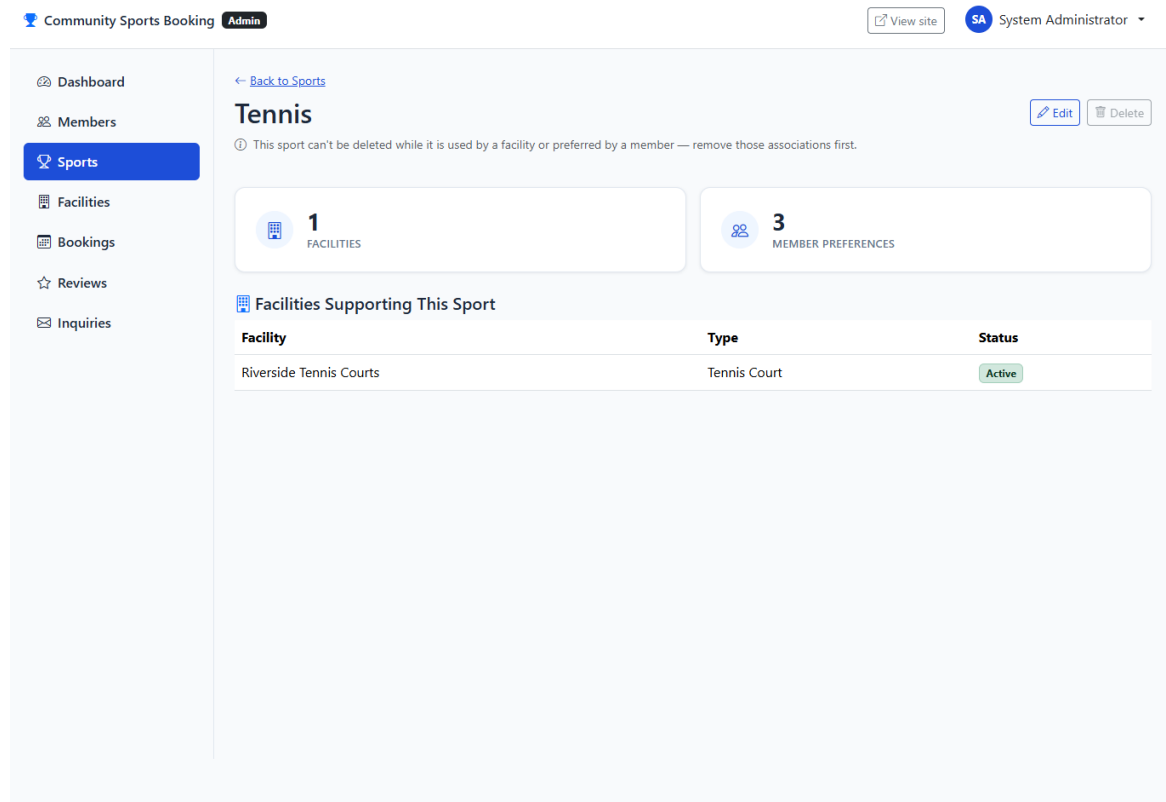
Sports list showing the new "Table Tennis" entry after creation.

Note: This sport was a disposable test entry and was removed after verification to keep the reference data unchanged for grading, in the same way the temporary test member and facility below were removed.

TC37 - Admin cannot delete a sport that is still in use

Test Case ID	TC37
Feature / Module	Sport Management (Admin)
Test Scenario	Admin cannot delete a sport that is still in use
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).
Test Steps	1. Open the details page for "Tennis" (a sport used by an active facility and preferred by members). 2. Observe the Delete control.
Test Input / Data	Sport = Tennis.
Expected Result	The Delete control is disabled and an explanation is shown that the sport cannot be deleted while it is used by a facility or preferred by a member.
Actual Result	Delete control disabled; on-page explanation confirms the sport is still in use.
Test Status	Pass

Evidence	Screenshot: TC37_Admin_Sport_Delete_Blocked.png
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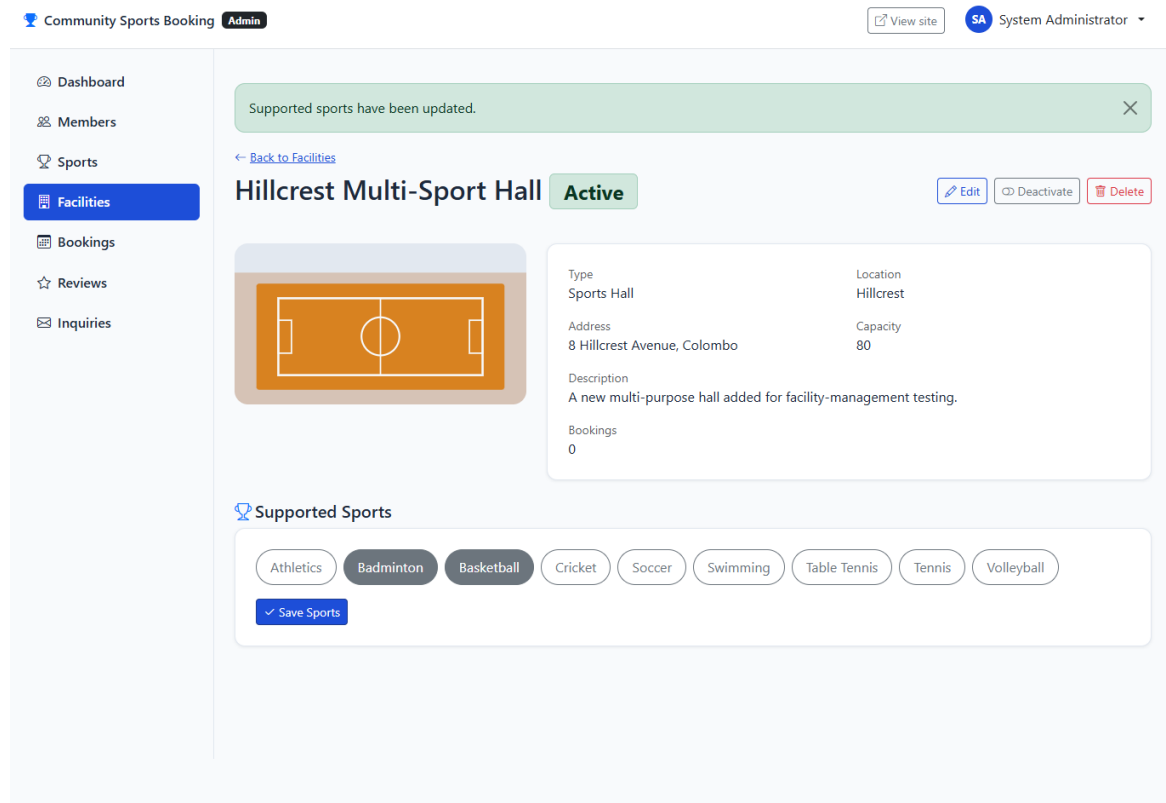


Sport details page showing the disabled Delete control and explanation for Tennis.

TC38 - Admin creates a facility and assigns supported sports

Test Case ID	TC38
Feature / Module	Facility Management (Admin)
Test Scenario	Admin creates a facility and assigns supported sports
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).
Test Steps	1. Open Admin > Facilities > Add Facility. 2. Fill in the facility details. 3. Save the facility. 4. On the facility's details page, select Basketball and Badminton as supported sports. 5. Click "Save Sports".
Test Input / Data	Facility Name = "Hillcrest Multi-Sport Hall", Type = "Sports Hall", Location = "Hillcrest", City = "Colombo", Address = "8 Hillcrest Avenue", Capacity = 80, Supported Sports = Basketball, Badminton.
Expected Result	The facility is created and the selected sports are saved as its supported sports.
Actual Result	Facility created; Basketball and Badminton assigned as supported sports.

Test Status	Pass
Evidence	Screenshot: TC38_Admin_Facility_Created_Sports_Assigned.png



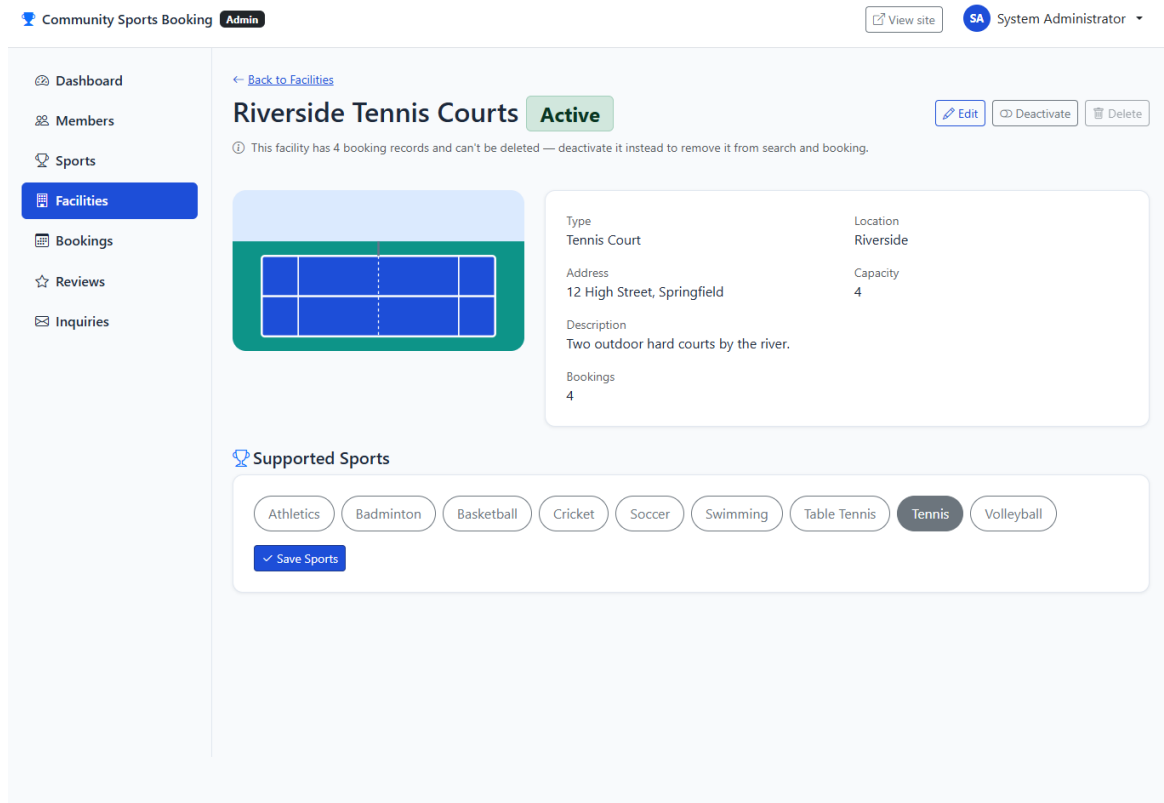
New facility details page showing Basketball and Badminton assigned as supported sports.

Note: This facility was a disposable test entry and was removed after verification, for the same reason described for TC36.

TC39 - Admin cannot delete a facility with existing booking history

Test Case ID	TC39
Feature / Module	Facility Management (Admin)
Test Scenario	Admin cannot delete a facility with existing booking history
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).
Test Steps	1. Open the details page for "Riverside Tennis Courts" (a facility with existing bookings). 2. Observe the Delete control.
Test Input / Data	Facility = Riverside Tennis Courts.
Expected Result	The Delete control is disabled and an explanation is shown that the facility has booking records that must be preserved, and should be deactivated instead.

Actual Result	Delete control disabled; on-page explanation confirms booking history must be preserved.
Test Status	Pass
Evidence	Screenshot: TC39_Admin_Facility_Delete_Blocked.png

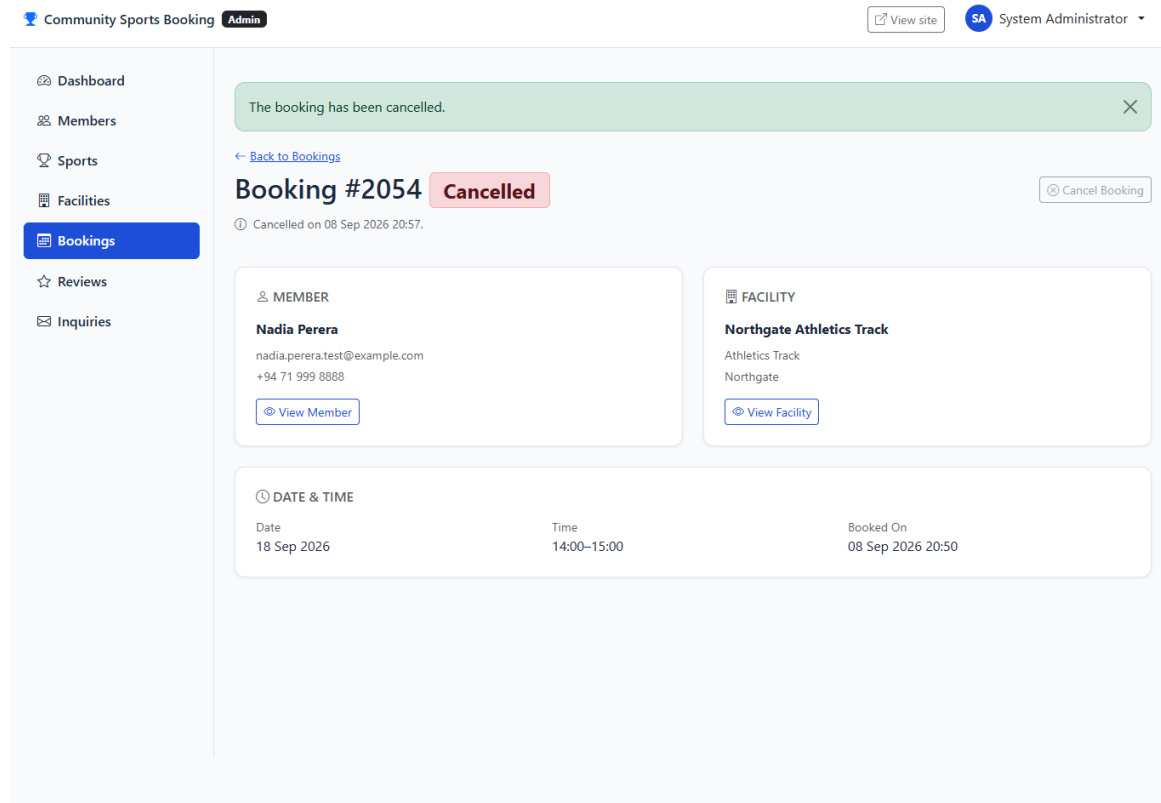


Facility details page showing the disabled Delete control and explanation for Riverside Tennis Courts.

TC40 - Admin filters bookings by member and cancels one

Test Case ID	TC40
Feature / Module	Booking Management (Admin)
Test Scenario	Admin filters bookings by member and cancels one
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).
Test Steps	1. Open Admin > Bookings. 2. Filter by member name "Nadia". 3. Click "Cancel" on the matching booking. 4. Confirm the cancellation.
Test Input / Data	Member filter = "Nadia".
Expected Result	The filtered list shows the matching booking. After cancelling, the booking status changes to Cancelled and the cancellation date is recorded.

Actual Result	Filtered list showed the matching booking; cancellation succeeded and status updated.
Test Status	Pass
Evidence	Screenshot: TC40_Admin_Booking_Filter_Cancel.png

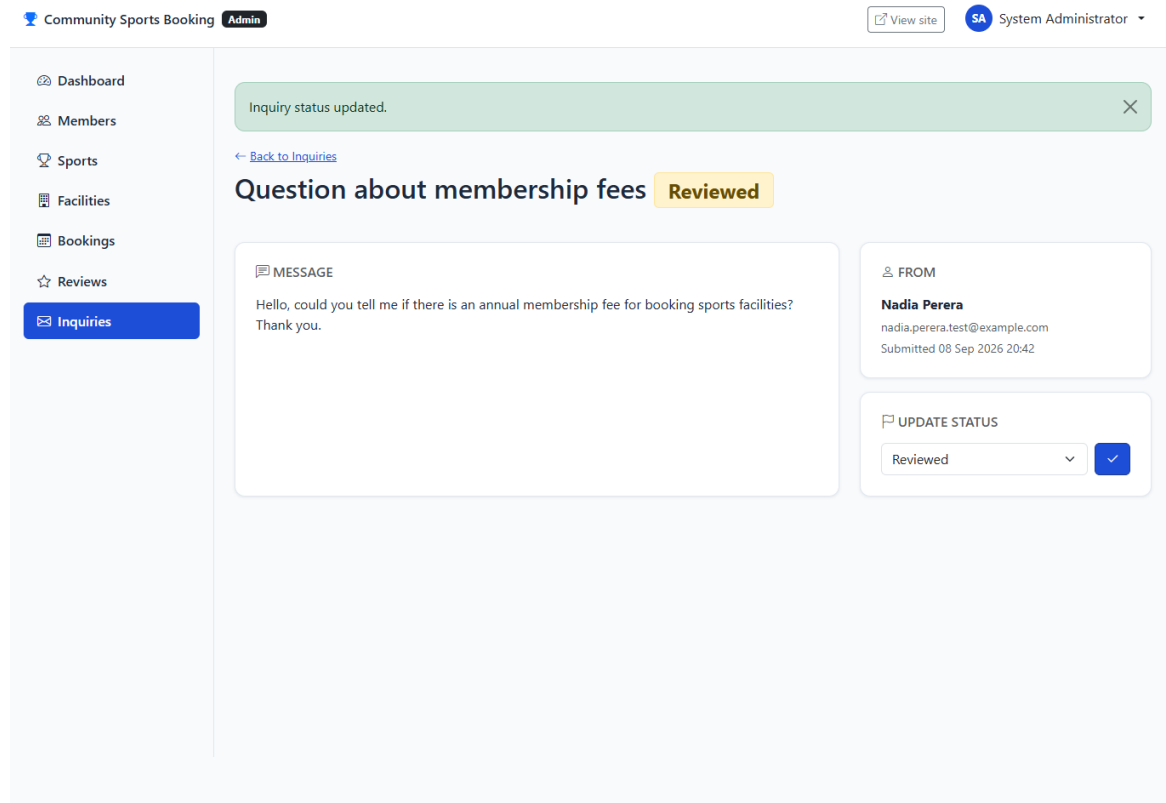


Booking details page showing the Cancelled status and cancellation confirmation.

TC41 - Admin reviews and updates the status of a guest inquiry

Test Case ID	TC41
Feature / Module	Inquiry Management (Admin)
Test Scenario	Admin reviews and updates the status of a guest inquiry
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area). The inquiry submitted in TC08 exists with Status = "New".
Test Steps	1. Open Admin > Inquiries. 2. Search for "membership". 3. Open the matching inquiry. 4. Change its status to "Reviewed". 5. Save the change.
Test Input / Data	Inquiry = "Question about membership fees", New Status = "Reviewed".
Expected Result	The inquiry status changes from New to Reviewed and a confirmation message is shown.

Actual Result	Inquiry found by search; status changed from New to Reviewed.
Test Status	Pass
Evidence	Screenshot: TC41_Admin_Inquiry_Status_Updated.png



Inquiry details page showing the status updated to Reviewed.

TC42 - Admin views all member-submitted reviews

Test Case ID	TC42
Feature / Module	Review Management (Admin)
Test Scenario	Admin views all member-submitted reviews
Preconditions	The application is running. The user is logged in as the System Administrator (Admin area).
Test Steps	1. Open Admin > Reviews.
Test Input / Data	None.
Expected Result	The full list of reviews across all facilities is shown, including the review submitted in TC26.
Actual Result	Review list includes Nadia's new review for Central Community Pool alongside seeded reviews.
Test Status	Pass
Evidence	Screenshot: TC42_Admin_Review_List.png

Community Sports Booking
Admin
View site
SA System Administrator

Dashboard
Members
Sports
Facilities
Bookings
Reviews
Inquiries

Reviews

3 reviews matching the current filters.

Member
Name or email

Facility
Facility name

Rating
All

Search
Clear

Central Community Pool · Nadia Perera
08 Sep 2026

★★★★☆

Clean and well-maintained pool with friendly staff. Would visit again.

[View details →](#)

Riverside Tennis Courts · Alice Johnson
30 Aug 2026

★★★★★

Excellent courts, well maintained and the surface was in great condition.

[View details →](#)

Central Community Pool · Ben Carter
30 Aug 2026

★★★★☆

Clean facility, but quite crowded during the afternoon session.

[View details →](#)

Admin review list showing all member-submitted reviews.

16. Test Summary

Test Group	Test Cases	Passed	Failed
Guest Module	12 (TC01-TC12)	12	0
Member Module	16 (TC13-TC28)	16	0
Administrator Module	14 (TC29-TC42)	14	0
Manual functional testing total	42	42	0
Automated xUnit test suite	185	185	0

All 42 manually executed test cases passed, and all 185 automated tests passed. No test case in either set was marked as passed without actually being run; where a result depended on a specific piece of prepared data (for example, a completed booking for the review tests), this is stated in the test case's preconditions rather than left unexplained.

17. Limitations

- The relational (physical) model image in Section 12.1 was engineered inside Oracle SQL Developer Data Modeler and should be checked against the actual SQL Server tables in Section 12.2 before submission. On a visual comparison, a few details in that image do not currently match the implemented database: the Booking table appears with a composite key (BookingId, MemberId, FacilityId) rather than the single BookingId key the database actually uses, the Member and Facility tables show incoming foreign key columns from Booking and MemberSport/FacilitySport (the reverse of the intended direction), and an extra

"Facility_and_Booking" table appears that has no equivalent in the real schema. These look like they come from how a couple of relationships were drawn in the logical model before the relational model was engineered from it, and are worth reviewing in Data Modeler alongside the table structures in Section 12.2, which do match the real database exactly.

- The login lockout described in Section 11.2 is held in the web server's memory. It resets if the application is restarted, and would not be shared automatically across more than one running copy of the application.
- There is no "forgot password" or password reset feature. A member who forgets their password cannot currently recover their account without administrator help.
- There is no email notification when a booking is made, cancelled, or when an inquiry is answered.
- Search results for facilities and reviews are not paginated. This is not a problem at the current amount of sample data, but a much larger number of facilities or reviews would benefit from paging.
- An administrator can view submitted reviews but cannot remove or edit one; review moderation was not part of the requested scope.
- While capturing evidence for TC33, a real inconsistency was noticed: the Admin > Members > member details page shows every booking as either "Upcoming" or "Completed" only. It does not yet check whether a booking has been cancelled, so a cancelled booking can appear there as "Upcoming" (or "Completed") instead of "Cancelled". The correct Cancelled status is shown properly everywhere else this information appears - the member's own My Bookings page, and the Admin > Bookings list and details pages - so this is a display gap on that one specific page rather than a problem with cancellation itself.
- The system was tested on a single local development environment. Its behaviour when deployed to a live hosting environment, or under many simultaneous users, has not been separately tested.

18. Application Source Code and Project Access

The complete Community Sports Facilities Booking System project, including the ASP.NET Core MVC application, source code, database scripts, and other required project files, is available through the following Google Drive link.

Google Drive Link:

[PASTE GOOGLE DRIVE LINK HERE]

The Google Drive folder contains the complete project files required to run and review the Community Sports Facilities Booking System. The project includes the ASP.NET Core MVC application source code, SQL Server database scripts, configuration files, and other supporting files required for the system.

19. Conclusion

The Community Sports Facilities Booking System meets the requirements set out in the case study. It provides a working home page, a complete set of Guest features (browsing, restricted search, review search, registration and inquiries), a complete set of Member features (search with availability, booking with genuine double-booking protection, booking history, cancellation, reviews and profile management), and an Administrator area for managing members, sports, facilities, bookings, reviews and inquiries.

The database was designed with normalisation in mind, and important rules - such as one review per booking, and no overlapping bookings for the same facility - are enforced by the database itself rather than only by the application. The finished system was tested with both an automated test suite and genuine manual testing of the running application, and every result recorded in this document reflects what was actually observed when the tests were run.

Overall, the project demonstrates a complete, working example of a database-backed web application, built using the technologies required by the coursework, and documents its design, features and testing honestly and in full.